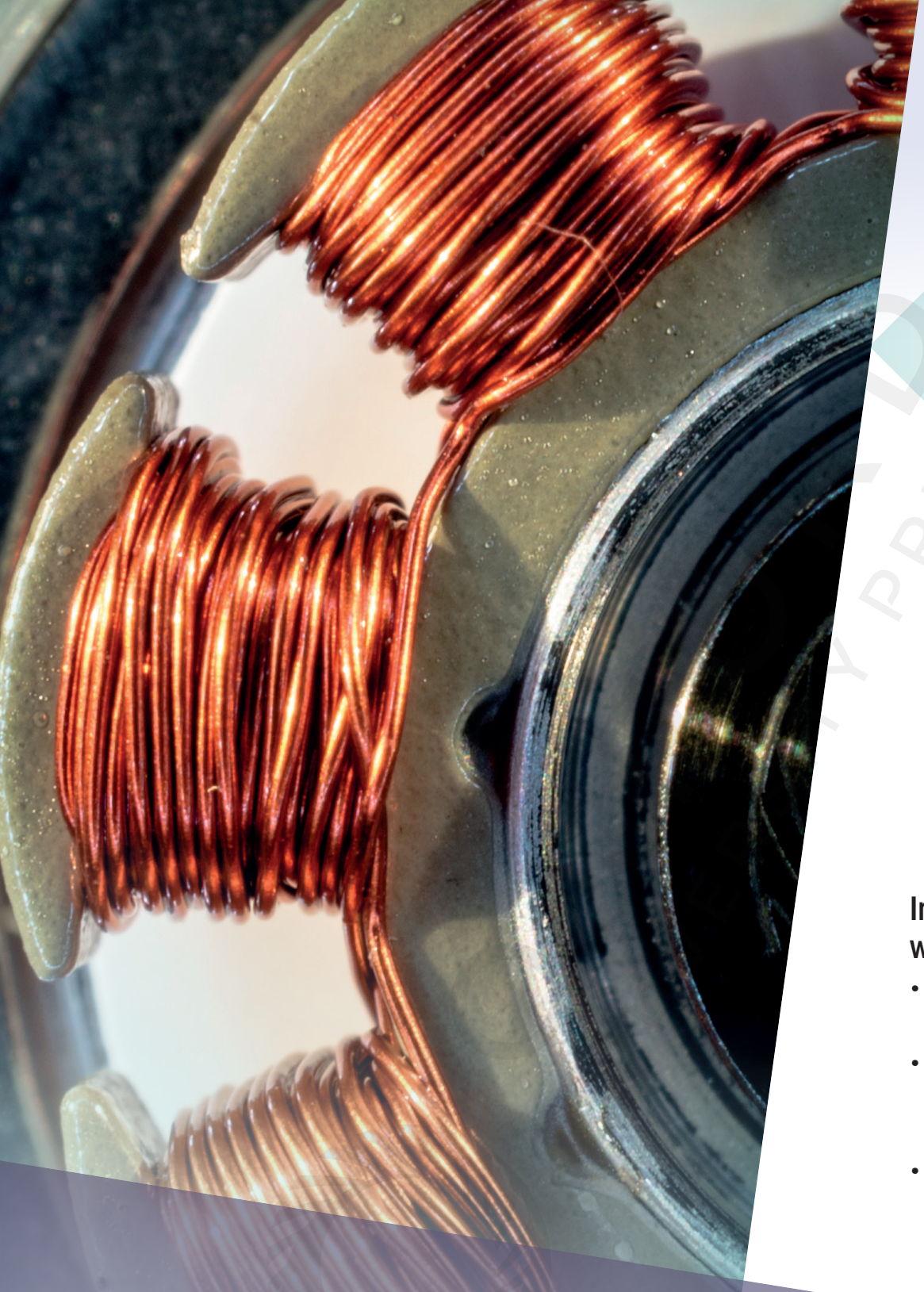


7

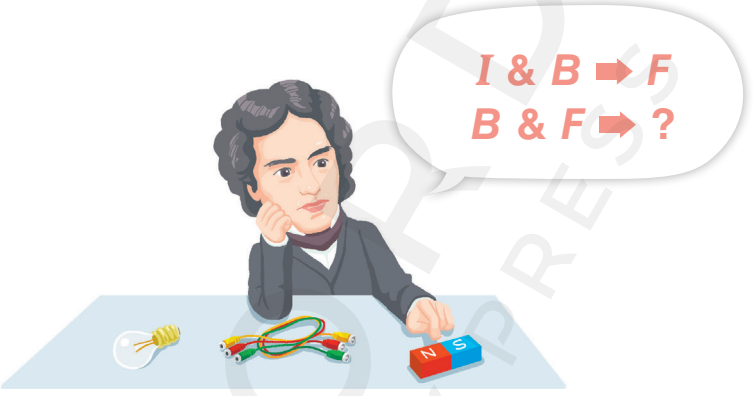
# Electromagnetic Induction

**In this chapter, we will learn about**

- the generation of electricity from magnetism
- the working principle of simple d.c. and a.c. generators
- other applications of electromagnetic induction

# 7.1 Generation of current with a magnetic field

You use electricity every day. Have you ever wondered how electricity is generated?




**DSE goal**

Examine induced e.m.f. resulting from a moving conductor in a steady magnetic field or a stationary conductor in a changing magnetic field.

**DSE exam**

21(1A)Q29

**Flipped classroom**

Watch a video and answer the questions.

The galvanometer can detect a very small current. ▶

**Video**

→ This video demonstrates Expt 7a.

→ This is a Video Quiz about this experiment.

## 1 Induced e.m.f. and current

### a Relative motion of conductor and magnet

#### Experiment 7a / Coil experiencing a change in magnetic field

1 Wind a 10-turn coil and connect it to a light-beam galvanometer.

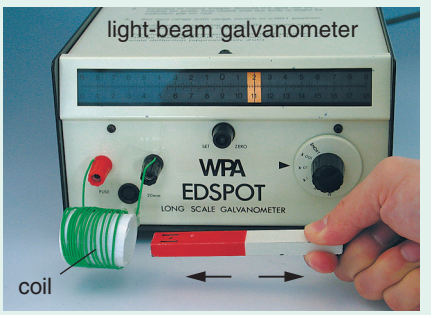
2 Observe the details of any deflection of the light beam when

(a) the magnet is pushed towards the coil;

(b) both the magnet and the coil are held stationary;

(c) the magnet is pulled away from the coil;

(d) the coil is pushed towards and then pulled away from the magnet.




**Discussion**

What is in common for a current to be generated in the above cases?

In Experiment 7a, the light beam of the galvanometer is deflected only when there is relative motion between the magnet and the coil. The direction of deflection changes according to the direction of this relative motion (Fig 7.1a on p.269).

(i) (ii)

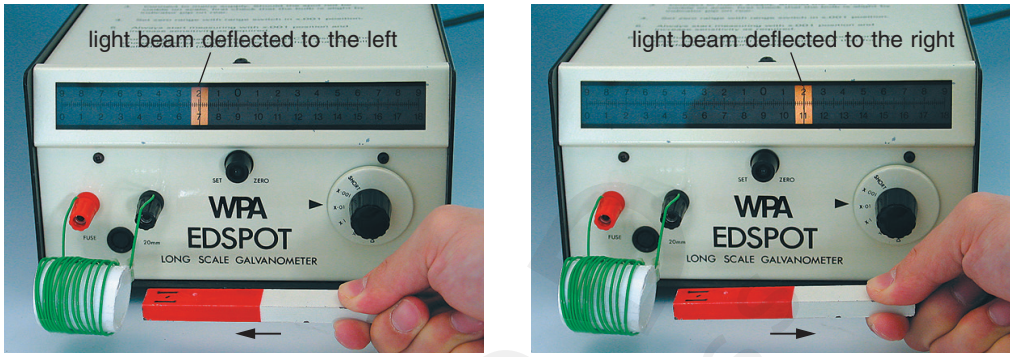



**Fig 7.1a** *The direction of current changes with the direction of the relative motion.*

This result suggests that an **e.m.f.** is produced in the coil when there is a relative motion between the coil and the magnet. Such an e.m.f. is called an **induced e.m.f.** and this effect is called **electromagnetic induction**.

In fact, the coil experiences a change in magnetic field due to the relative motion (Fig 7.1b). This change produces the induced e.m.f.




**Fig 7.1b** *Magnetic field experienced by the coil becomes stronger as the magnet is closer.*

If the coil is connected in a **closed circuit** as in the experiment, the induced e.m.f. will drive an **induced current** to flow in the circuit.

Stress that there is no induced current if the circuit is open. However, an e.m.f. is always induced when the coil experiences a change in magnetic field.

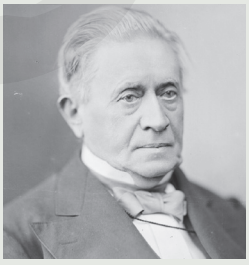
## Know more

### Michael Faraday (1791–1867)

*Michael Faraday* was an English scientist. Besides discovering electromagnetic induction, he also invented the dynamo and the transformer (see Chapter 8).

▸ p.319





### Joseph Henry (1797–1878)

*Joseph Henry* was an American scientist. He observed electromagnetic induction a few months before Faraday did, but published the discovery later. He improved the design of electromagnets and created the prototype of a.c. motors.

induced e.m.f. 感生電動勢　electromagnetic induction 電磁感應　induced current 感生電流
Michael Faraday 法拉第　Joseph Henry 亨利

Video

→ This video demonstrates Expt 7b.

→ This is a Video Quiz about this experiment.

## Experiment 7b / Induced current

1 Connect a wire to a light-beam galvanometer.

2 Observe the deflection of the light beam when the wire is moved between a pair of slab-shaped magnets
   (a) vertically up and down,
   (b) forwards and backwards (Fig a),
   (c) left and right.

3 Repeat step 2(a) by
   (a) moving the wire at different speeds,
   (b) using stronger magnets,
   (c) replacing the wire with a coil of several turns (Fig b).

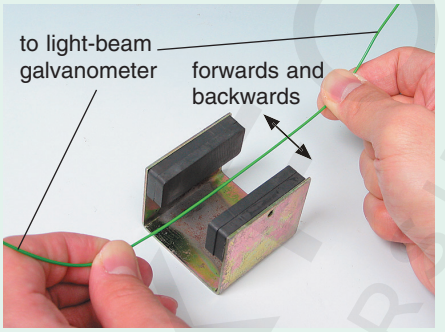



*Fig a*

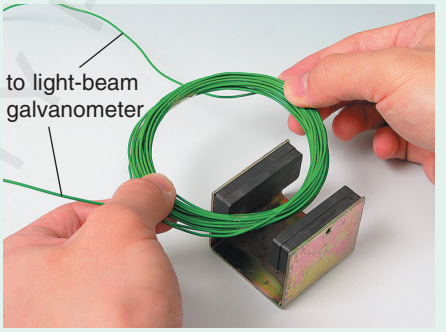



*Fig b*

### Discussion

1 How should the wire move in the magnetic field to induce an e.m.f.?

2 How can the magnitude of the induced e.m.f. be increased?

## b Faraday's law of electromagnetic induction

**Cross-link**
You will learn how to determine the '+' and '–' signs on p.275. ▶

In Experiment 7b, an e.m.f. is induced when the wire moves vertically, i.e. when the wire cuts through the magnetic field lines (Fig 7.1c). No e.m.f. is induced when the wire does not cut through the field lines (Fig 7.1d).

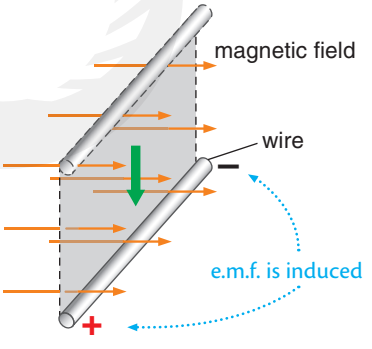



***Fig 7.1c*** *The wire cuts the field lines.*

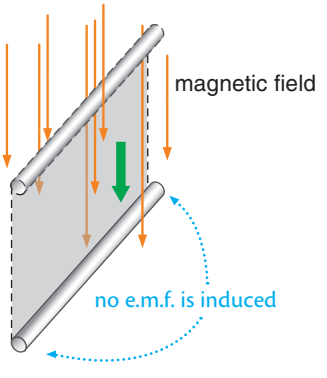



***Fig 7.1d*** *The wire does not cut the field lines.*

From the results of Experiments 7a and 7b, we can conclude that

**An e.m.f. is induced if**

**1 the magnetic field through a coil is changing, or**

**2 a conductor is cutting through magnetic field lines.**

**The magnitude of the induced e.m.f. can be increased by**

**1 moving the conductor or the magnet faster ($v \uparrow$),**

**2 using a stronger magnet ($B \uparrow$),**

**3 using a coil with more turns to increase the length of the conductor within the magnetic field ($N \uparrow$ or $L \uparrow$).**

**Caution**
A bigger magnet is not necessarily stronger.

**Caution**
It is the number of turns (total length of wire in the field) that matters, not the number of turns per unit length.

These results are summarized in **Faraday's law of electromagnetic induction**:

**The magnitude of the induced e.m.f. is directly proportional to the rate of change of the magnetic field passing through a coil, or the rate at which a conductor cuts through the magnetic field lines.**

**Cross-link**
Strictly speaking, the induced e.m.f. is equal to the rate of change of magnetic flux. The magnetic flux measures the amount of magnetic field lines crossing an area. This will be discussed in Ch 7.3.

## Quick check 1

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

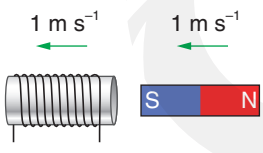
**1** *True or false:*

(a) An e.m.f. is induced in a coil if the magnetic field through the coil decreases. (T/F)

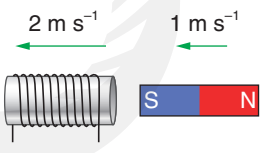
(b) An e.m.f. is induced whenever an object cuts through magnetic field lines. (T/F)

**2** Determine whether each of the following movements will result in an induced e.m.f. in the coil.

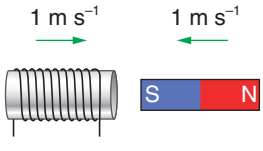
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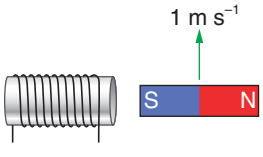

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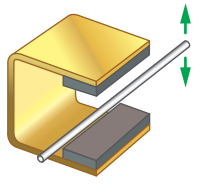



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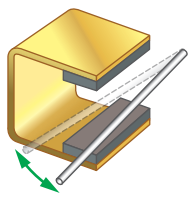



**3** Determine whether each of the following movements of the metal rod between two slab-shaped magnets will result in an induced e.m.f. across it.

☐



☐



**4** A magnet is moved away from a coil with speed $v$ and an e.m.f. is induced in the coil. Determine whether each of the following changes will result in a larger e.m.f..

☐ Use another coil with more turns.

☐ Move the magnet faster.

☐ Move both the magnet and the coil away from each other with speed $v$.

☐ Use a stronger magnet.

**DSE goal**

Apply Lenz's law to determine the direction of induced e.m.f./ current.

**DSE exam**

12(1A)Q31, 32, (1B)Q10, 13(1A)Q29, 14(1A)Q27, 18(1B)Q9, 21(1A)Q26

**Simulation**

→ This simulation shows how the current is induced when a magnet moves in and out of a coil. It is available on OUP web in PC version (.exe) only.

# 2 Direction of induced current and e.m.f.

## a Lenz's law

Consider a coil connected in a closed circuit (Fig 7.1e). When the N-pole of a magnet is moved towards the coil, a current is induced in the coil. The coil becomes an electromagnet. So, what is the magnetic polarity at the end *X* of the coil?

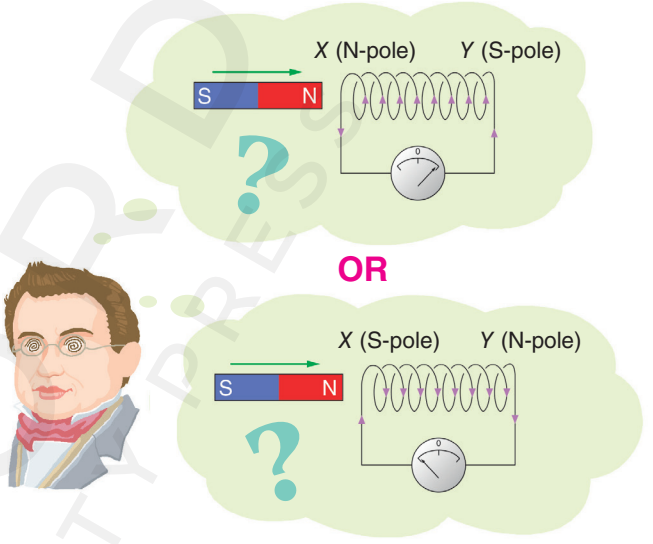



***Fig 7.1e*** *What is the polarity of X when an N-pole is approaching?*

If *X* were the S-pole, the magnet would be attracted towards the coil and would speed up, thus giving a larger induced current. This means that by giving the magnet a little kinetic energy, more and more kinetic energy and electrical energy would be resulted out of nothing. This is obviously **impossible**. Therefore, *X* must become the N-pole and repel the magnet.

This violates the law of conservation of energy. ▶

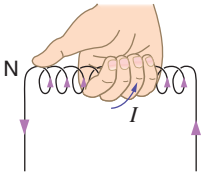
*Heinrich Lenz* discovered this phenomenon in 1834 and summarized his finding, which was later known as **Lenz's law**:

> **The induced current (or e.m.f.) is in the direction such that it opposes (or tends to oppose) the change causing it.**

Therefore, when the N-pole of a magnet is approaching a coil, the induced current will flow in a way to make end *X* an N-pole to oppose the motion of the magnet (Fig 7.1f). Conversely, if the N-pole is leaving the coil, *X* will become an S-pole to oppose the motion.

**Cross-link**

The direction of the induced current can be found using the right-hand grip rule. See Ch 5 (p.197).



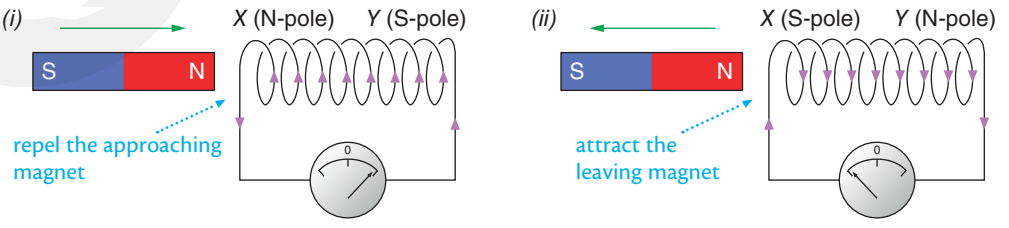




***Fig 7.1f*** *The induced current opposes the motion of the magnet (i) towards the coil, (ii) away from the coil.*

Heinrich Lenz 楞次　　Lenz's law 楞次定律

Lenz's law is an inevitable consequence of the conservation of energy. A current is induced in the coil to oppose the motion of the magnet. An external force is needed to keep the magnet moving, and the work done by this external force is converted into electrical energy.

## Know more

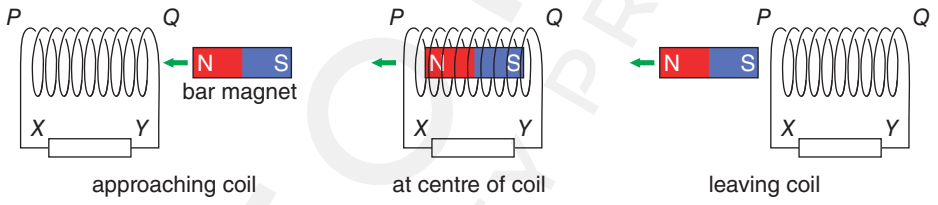
### Heinrich Lenz (1804–1865)

Heinrich Lenz was a Russian physicist. He is best known for formulating Lenz's law. He also discovered Joule's law ($P = I^2R$) independently of Joule.



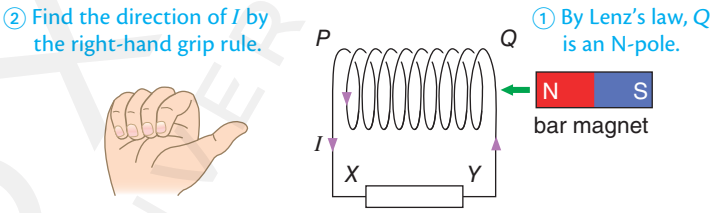
## Example 1 A magnet passing through a coil

A bar magnet passes through a coil as shown. Draw the direction of the induced current $I$ through the coil when the magnet is (a) approaching the coil, (b) at the centre of the coil, (c) leaving the coil. Hence, determine which end of the resistor, $X$ or $Y$, is at a higher potential.



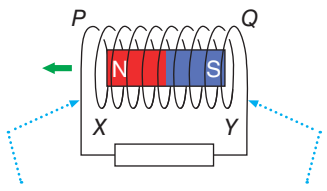

### Solution

(a)




End $X$ is at a higher potential.

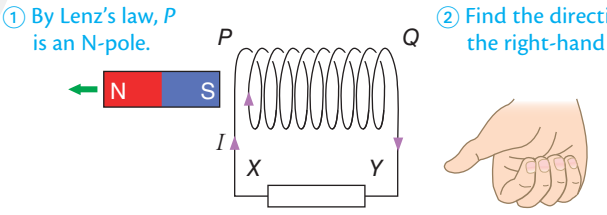
(b)




The e.m.f. induced at the two ends cancel each other out.

There is no induced current. $X$ and $Y$ are at the same potential.

(c)




End $Y$ is at a higher potential.

**Tip**

Current flows from a position of higher potential to a position of lower potential in the external circuit.

## Know more

### Shake-shake torch



You need not replace the battery in this torch. When you shake the torch, a magnet passes in and out of a coil. The induced current charges the battery.

**Video**

→ This video demonstrates how a shake-shake torch works.

In part (a) of Example 1, current flows from end *P* of the coil through the resistor and back to end *Q*. The coil acts like a battery with end *P* being the positive (+) terminal (Fig 7.1g). In other words, end *P* is at a higher potential than end *Q*.

If the resistor is removed, the circuit is **open**. **No current** will flow in the coil, but an **e.m.f. is still induced** across it, and end *P* is still at a higher potential than end *Q* (Fig 7.1h). Note that when there is **no current**, **no opposing force** acts on the magnet.

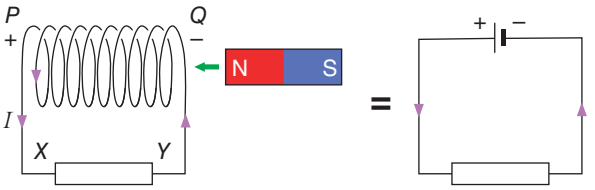



***Fig 7.1g*** *An e.m.f. is induced in the coil and the coil acts like a battery.*

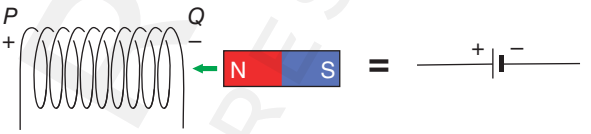
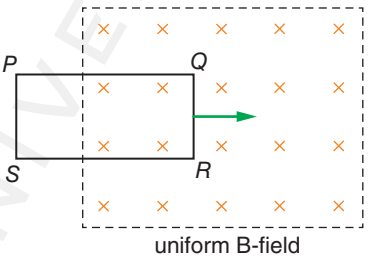



***Fig 7.1h*** *An e.m.f. is still induced in the coil even when the circuit is open.*

## Example 2 A coil entering a uniform magnetic field

A retangular metal coil *PQRS* is entering a uniform magnetic field which points into the page.




(a) Explain the direction of the induced current in the coil.

(b) Explain whether a current is induced when the coil is completely inside the field.

### Solution

(a) As the coil moves, it encloses more and more magnetic field lines that point into the page.

According to Lenz's law, to oppose this change, the induced current flows in a direction to produce field lines that point out of the page.

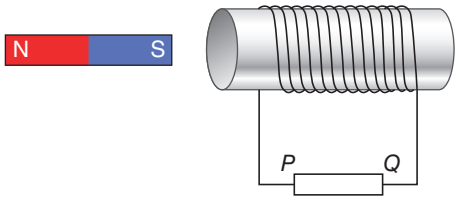
By the right-hand grip rule, the induced current flows anticlockwise.

(b) No current is induced in the coil. This is because the number of field lines enclosed by the coil remains unchanged.

## Quick check 2

Google Forms | Microsoft Forms → E-version in OUP Exercise Platform is also available.

**1**




In each of the following cases, (i) what is the direction of the induced current? (ii) Which end, *P* or *Q*, is at a higher potential?

(a) The magnet is moving to the left.

(b) The magnet is moving to the right.

(c) The coil is moving to the right.

**2** *True or false:*

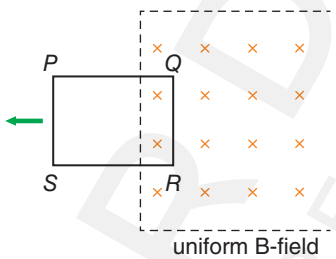
(a) When a magnet moves towards a coil which is connected to an open circuit, no e.m.f. is induced across the coil. (T/F)

(b) The induced current always opposes the change causing it. (T/F)

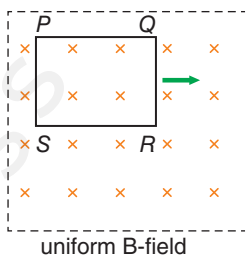
(c) Lenz's law can only be applied to closed circuits. (T/F)

**3** In each of the following cases, find the direction of the induced current in the coil *PQRS* when the coil moves as indicated.

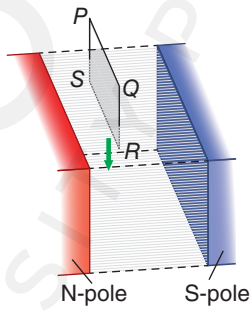
(a)




(b)




(c)




Remind Ss that this is a rule, not a physics principle, which helps them remember the directions of the force, magnetic field and induced current.

**Caution**

Do not mix it up with Fleming's left-hand rule, which deals with the magnetic force acting on a current in a magnetic field.

## b Fleming's right-hand rule

In Figure 7.1i, a straight conductor is moved upwards through a magnetic field. How can you find the direction of the induced current? **Fleming's right-hand rule** may help you to find out the answer.

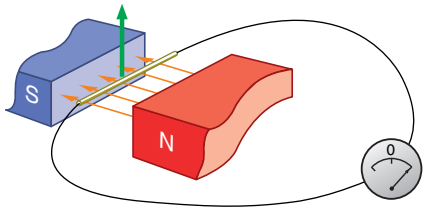
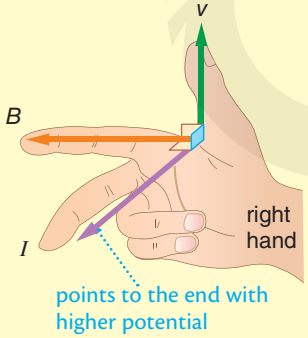



***Fig 7.1i*** *A conductor moving in a magnetic field.*

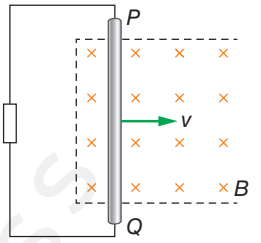



**Fleming's right-hand rule:**

1. **Hold the thumb and the first two fingers of the right hand at right angles.**
2. **The thumb gives the direction of motion of the conductor (*v*).**
3. **The first finger gives the direction of the magnetic field (*B*).**
4. **The second finger gives the direction of the induced current (*I*). It also points to the end of the conductor with a higher electric potential.**

## Example 3 A conductor cutting through a magnetic field

A straight conductor $PQ$ is connected to a resistor. It is moved to the right in a uniform magnetic field pointing into the page as shown.




(a) State the direction of the induced current through the conductor $PQ$.

(b) Which end, $P$ or $Q$, is at a higher potential?

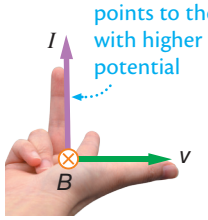
### Solution

(a) From $Q$ to $P$

(b) End $P$

**Tip**

By Fleming's right-hand rule, points to the end with higher potential




In this example, end $P$ is just like the positive (+) terminal of a battery (Fig 7.1j). If the resistor is removed, the circuit is **open** and **no current** will flow in $PQ$, but an e.m.f. is still induced across $PQ$.

May tell Ss that in case (ii), end $P$ becomes positively charged and end $Q$ becomes negatively charged. A more detailed discussion can be found on p.319 Know more.

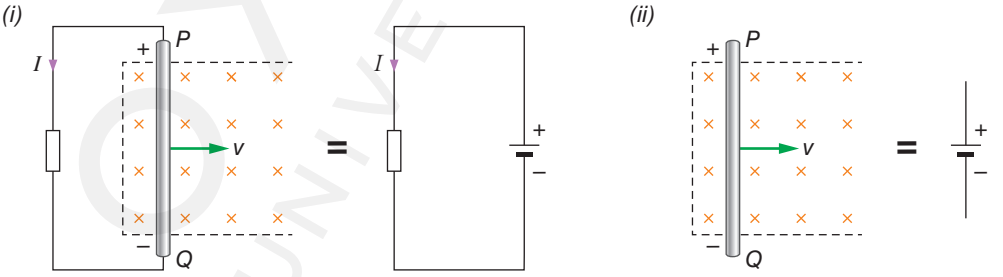



***Fig 7.1j*** *An e.m.f. is induced in (i) a closed circuit, (ii) an open circuit.*

Note that when the induced current flows from $Q$ to $P$, a magnetic force $F$ is produced. By Fleming's **left**-hand rule, this force points to the left (Fig 7.1k). This agrees with Lenz's law that the induced current flows to oppose the change (motion of $PQ$).

Try to redo Example 3 with Lenz's law and Example 2 with Fleming's right-hand rule.

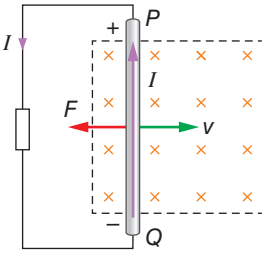


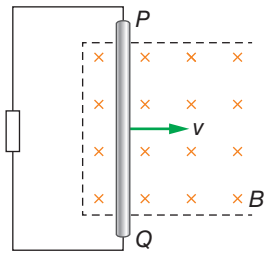

***Fig 7.1k*** *Magnetic force acting on PQ due to the induced current.*

## Quick check 3

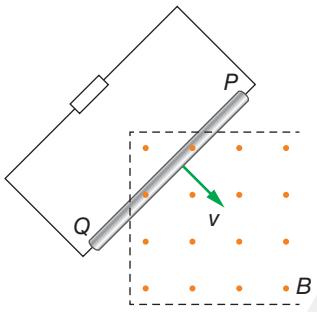
Google Forms Microsoft Forms → E-version in OUP Exercise Platform is also available.

**1** A metal rod *PQ* is moved in a uniform magnetic field *B* in the directions shown below.

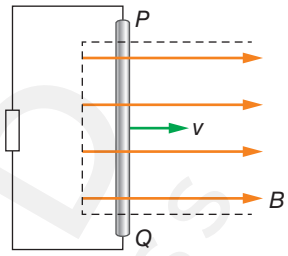
(a)




(b)




(c)



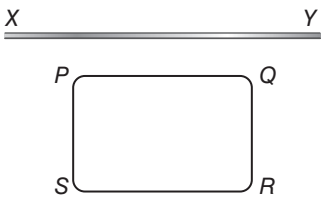

(i) What is the direction of the induced current through the rod?

(ii) Which end of the rod, *P* or *Q*, is at a higher potential?

☆ HKDSE 2014 Paper 1A Q27

## Example 4 A coil beside a straight wire

A coil *PQRS* is placed next to a long straight wire *XY* as shown.




Which of the following can generate a clockwise induced current in *PQRS*?

(1) A steady current flows from *Y* to *X*.

(2) A current flows from *X* to *Y* and is decreasing in magnitude.

(3) A current flows from *Y* to *X* and is increasing in magnitude.

A (1) only B (2) only

C (2) and (3) only D (1), (2) and (3)

### Solution

When the current is steady, the magnetic field through *PQRS* does not change, so no current is induced.

∴ (1) is incorrect.

When a current flows from *X* to *Y*, by the right-hand grip rule, the magnetic field points into the page through *PQRS*. When the current decreases, the magnetic field decreases. By Lenz's law, the induced current produces a magnetic field pointing into the page to oppose the change. By the right-hand grip rule again, the induced current flows clockwise.

∴ (2) is correct.

Similarly, when a current flows from *Y* to *X*, the magnetic field points out of the page through *PQRS*. When the current increases, the induced current produces a magnetic field pointing into the page to oppose the change.

∴ (3) is correct.

∴ The answer is C.

▶ Revision 7 Q10 (p.311)

By Fleming's left-hand rule, the magnetic force acting on *PQ* points towards *XY* while that on *SR* points away from *XY*. Since the magnetic field at *PQ* is stronger than that at *SR*, the magnetic force acting on *PQ* is stronger.

∴ *PQRS* is attracted towards *XY*.

*PQRS* is repelled by *XY*.

## Example 5 A bar magnet falling through a coil

A bar magnet with its N-pole pointing downwards is dropped and passes through a coil (Fig a). A voltage sensor is used to record the induced e.m.f. $\varepsilon$ in the coil. Figure b shows how $\varepsilon$ varies with time.

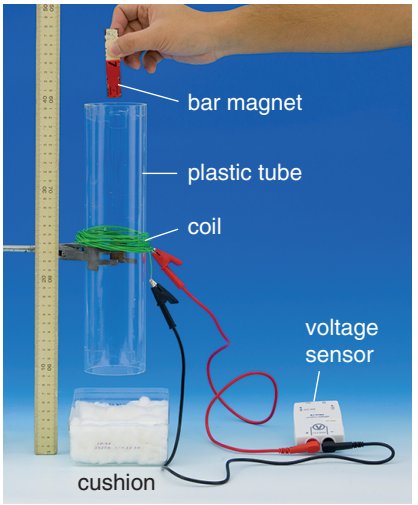



*Fig a*

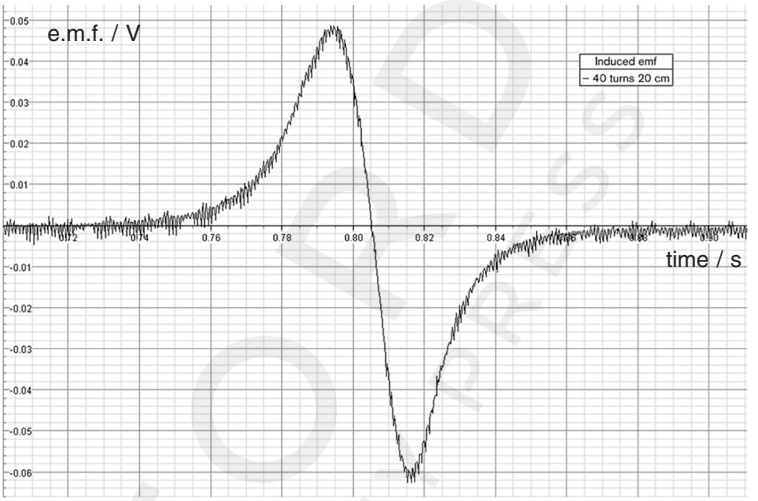



*Fig b*

(a) Explain why $\varepsilon$ changes sign in the process.

(b) Explain why the negative peak is larger in magnitude than the positive peak.

(c) State what happens to the graph if

(i) the polarity of the magnet is reversed;

(ii) the coil is replaced by another coil with more turns.

(d) If the coil is connected to a 1-Ω resistor instead of the voltage sensor, will the magnet fall faster or slower?

### Solution

(a) By Lenz's law, when the magnet is approaching the upper end of the coil, the e.m.f. induced tends to make the upper end an N-pole to oppose the approaching N-pole of the magnet. When the magnet is leaving the lower end of the coil, the induced e.m.f. tends to make the lower end an N-pole to attract the leaving S-pole of the magnet. Hence $\varepsilon$ changes sign.

(b) The magnet speeds up as it falls under gravity and therefore leaves the coil at a higher speed. This causes a higher rate of change in the magnetic flux through the coil and produces a larger magnitude of $\varepsilon$.

(c) (i) The graph becomes upside down.

(ii) Both the positive and negative peaks become larger in magnitude.

(d) Since the resistance of the circuit will decrease, the induced current will increase. The magnet will be opposed by larger magnetic forces and will fall slower.

▶ Revision 7 Q4 (p.310)

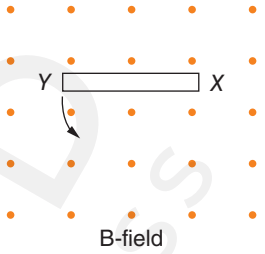
☆ HKDSE 2013 Paper 1A Q29, 2021 Paper 1A Q26

→ This video demonstrates how this question is solved.

Video

## DSE practice 1 | A metal rod moving in a magnetic field

A metal rod $XY$ is rotated about end $X$ in a uniform magnetic field pointing out of the page.




Which of the following statements about the rod is/are correct?

(1) End $X$ is at a higher potential than end $Y$.

(2) The rod experiences a magnetic force that opposes its motion.

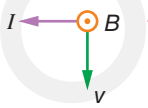
(3) The induced e.m.f. across the rod will be larger if the rod is longer.

A (1) only B (3) only

C (1) and (2) only D (2) and (3) only

### Solution

The rod (except end $X$) moves downwards. By Fleming's right-hand rule, end $Y$ is at a higher potential than end $X$.




**Common mistake** ⚠
Students may not know how to determine which end is at a higher potential.

$\therefore$ (1) is incorrect.

There is no induced current, hence no opposing force, since the rod is not in a closed circuit.

**Common mistake** ⚠
Students may not notice this.

$\therefore$ (2) is incorrect.

The induced e.m.f. increases when the length of the conductor cutting through the magnetic field increases.

$\therefore$ (3) is correct.

$\therefore$ The answer is B.

▶ Revision 7 Q16 (p.313)

## Exercise 7.1

★1 A bar magnet moves with a speed $v$ through a solenoid connected to a galvanometer. At the instant shown in Figure a, the magnet is leaving the solenoid.

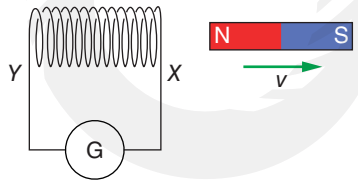



*Fig a*

Which of the following statements is/are correct?

(1) An N-pole is induced at $X$ at the instant shown.

(2) The reading of the galvanometer is 0 when the magnet is at the mid-point of the solenoid.

(3) The magnetic force acting on the solenoid points to the right no matter when the magnet is approaching or leaving it.

A (1) only B (2) only

C (1) and (3) only D (2) and (3) only

★2 A metal rod is moving and a current is induced in it. What can we say about the rod?

(1) The rod is part of a closed circuit.

(2) The magnetic force experienced by the rod is in opposite direction to the direction of motion of the rod.

(3) The rod is cutting through magnetic field lines.

A (1) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

*3 An aluminium ring is dropped from rest over a bar magnet from $P$ to $Q$ (Fig b).

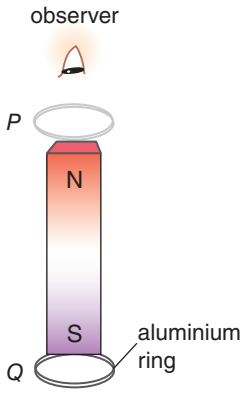



Fig b

Which of the following statements is/are correct?

(1) From the observer's view, the induced current flows in a clockwise direction round the ring at $P$.

(2) The ring is repelled by the magnet when it is at $Q$.

(3) The ring can reach a higher speed at $Q$ if it falls without the bar magnet.

A (1) only B (2) only

C (1) and (3) only D (2) and (3) only

*4

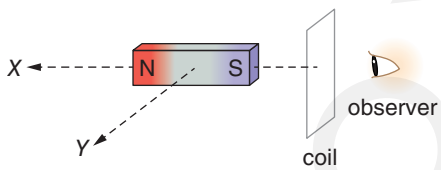



Fig c

State the direction of the current induced in the coil as seen by the observer in Figure c if the magnet moves

(a) along direction $X$,

(b) along direction $Y$.

*5 A magnet is suspended by an elastic string above a solenoid (Fig d). The magnet is pulled downwards slightly and then released so that it oscillates vertically.

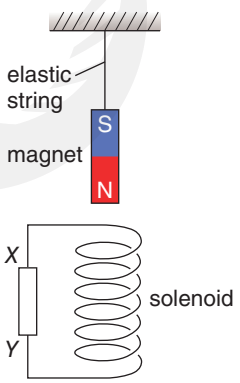



Fig d

(a) Describe the direction of the induced current flowing through resistor $XY$ as the magnet oscillates.

(b) Explain why the magnet stops oscillating quickly.

*6 A metal rod $PQ$ is falling in a uniform magnetic field (Fig e).

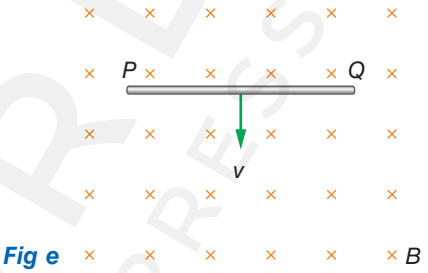



Fig e

(a) Which end, $P$ or $Q$, is at a higher potential?

(b) Is the acceleration of the rod higher than, equal to or lower than the acceleration due to gravity? Explain briefly.

*7 Paul sets up the apparatus as shown (Fig f). He drops a bar magnet from a height through the coil and the light beam is deflected.

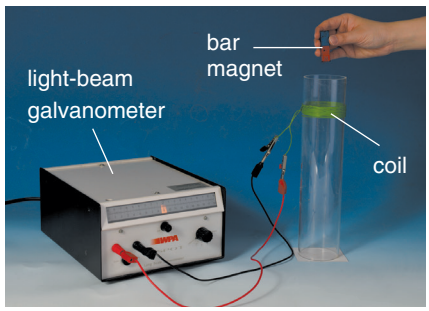



Fig f

Suggest two methods to increase the deflection of the light beam without changing the setting of the galvanometer.

*8 A 1100-turn coil is connected to a galvanometer **without** any magnets placed near it (Fig g). When the coil is flipped as shown in Figure g, a current is detected. Why?

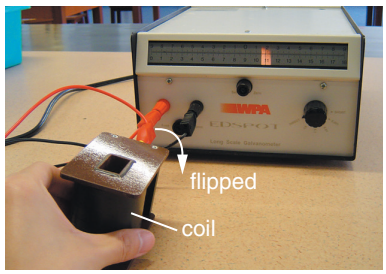



Fig g

# 7.2 Applications of electromagnetic induction

You may have seen similar devices on bicycles (Fig 7.2a). They are **generators** (or *dynamos*).




*Fig 7.2a* Different bicycle dynamos.

**Cross-link**
You learned the motor in Ch 6.2. It converts electrical energy into kinetic energy.

A generator is an important application of electromagnetic induction. It converts kinetic energy into electrical energy, just like a motor in reverse. Let us learn about its working principle.

**DSE goal**
Describe the structures of simple d.c. and a.c. generators and how they work.

**DSE exam**
15(1A)Q24, 22(1A)Q28

**Simulation**

→ The simulation 'Induced current in a rotating coil' shows how a coil rotating in a magnetic field produces an induced current in the coil.

→ The simulation 'Simple d.c. generator' shows how the simple d.c. generator works, and how the d.c. produced varies with time as the coil rotates.

## 1 Generators

### a Simple d.c. generators

A simple **d.c. generator** is like a simple d.c. motor working in reverse (Fig 7.2b). It has a coil situated between two opposite magnetic poles. The ends of the coil are fixed to a commutator, which is connected to an external circuit via a pair of carbon brushes. Unlike a simple d.c motor, it serves as a power source driving a load (the bulb) in the external circuit.

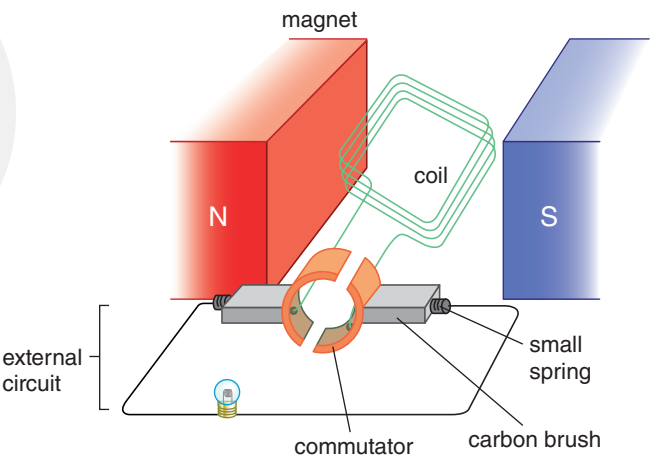



*Fig 7.2b* A simple d.c. generator.

generator/dynamo 發電機　　d.c. generator 直流發電機

When the coil is rotated by an **external force**, it cuts through the magnetic field and a current $I$ is induced (Fig 7.2c).

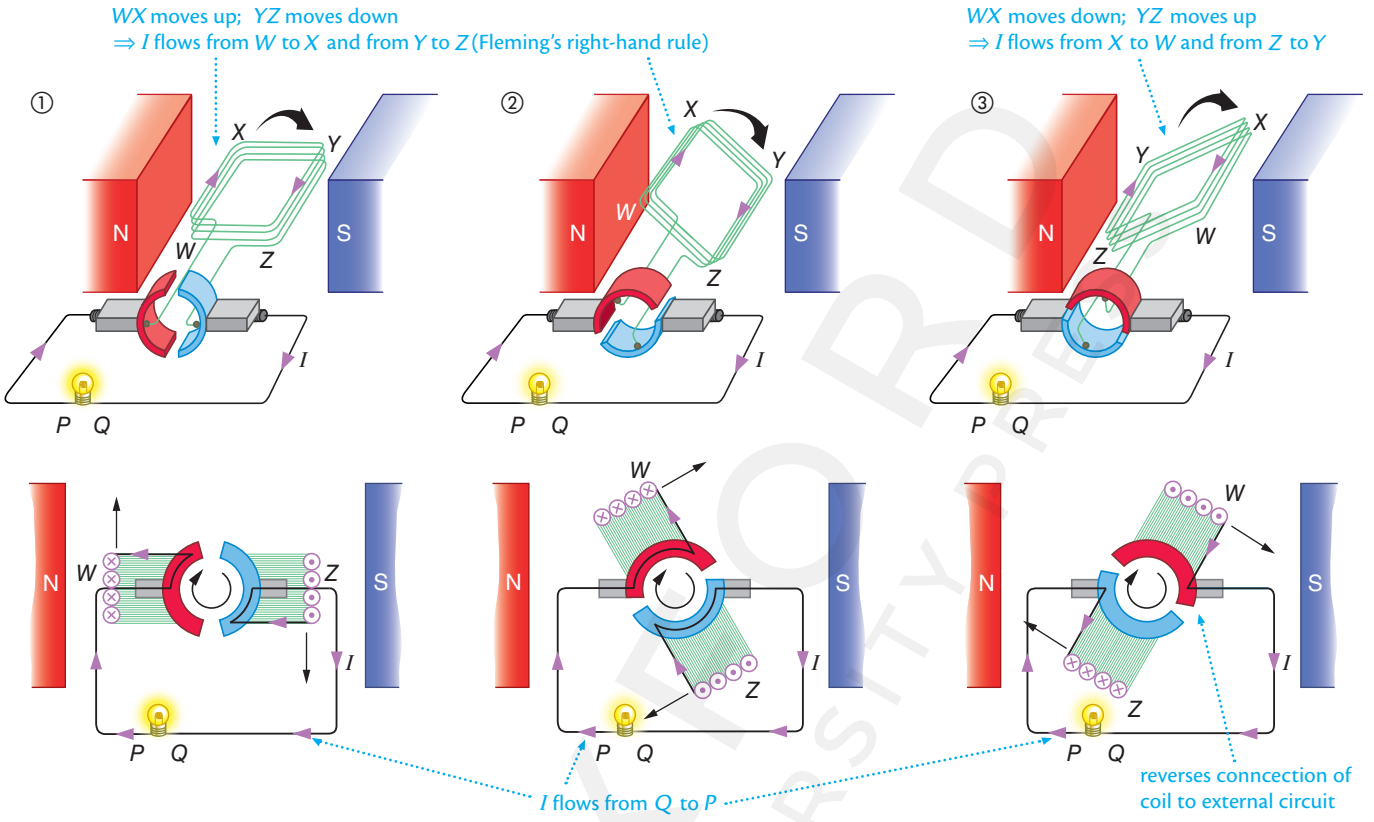



***Fig 7.2c*** *How a simple d.c. generator works.*

When the coil rotates, the rate at which sides $WX$ and $YZ$ cut the field lines varies, so the induced current varies. Figure 7.2d shows how the current through the **external circuit** varies in one complete rotation of the coil. The current does not change its direction, so it is a d.c.

The current in the coil is an a.c., but the current in the external circuit is a d.c.

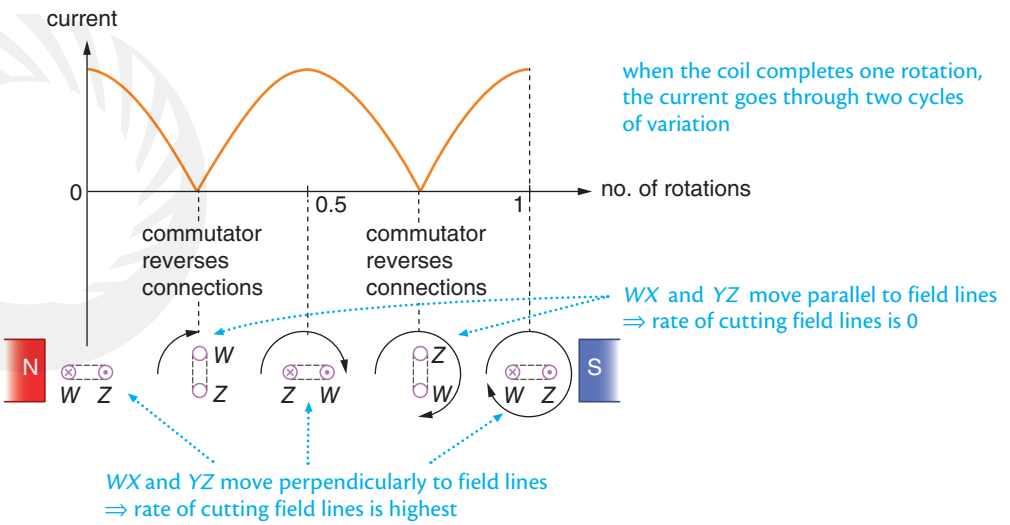



***Fig 7.2d*** *Current produced by a d.c. generator.*

Simulation

→ This simulation shows how a simple a.c. generator works, and how the a.c. produced varies with time as the coil rotates.

## b Simple a.c. generators

A simple d.c. generator becomes a simple **a.c. generator** if the commutator is replaced by a pair of **slip rings**. Each end of the coil is fixed to one ring. The rings are connected to the external circuit via carbon brushes. Figure 7.2e explains how an a.c. generator works.

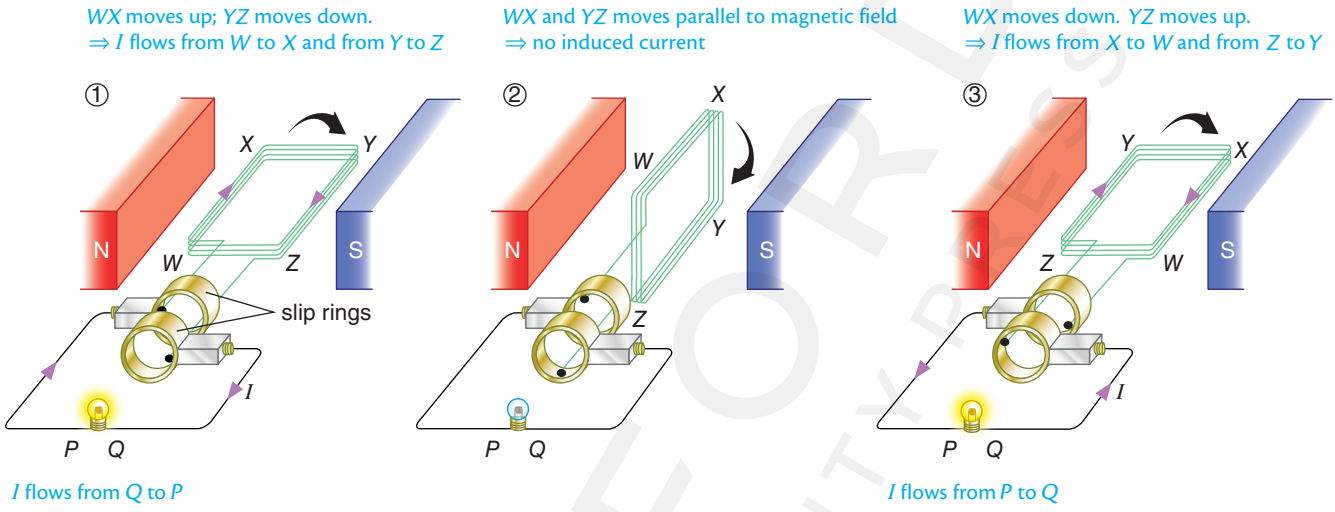



*Fig 7.2e* How a simple a.c. generator works.

The direction of current in the coil and the external circuit reverses whenever the coil passes the vertical position. Figure 7.2f shows how the current in the **external circuit** varies in one complete rotation of the coil.

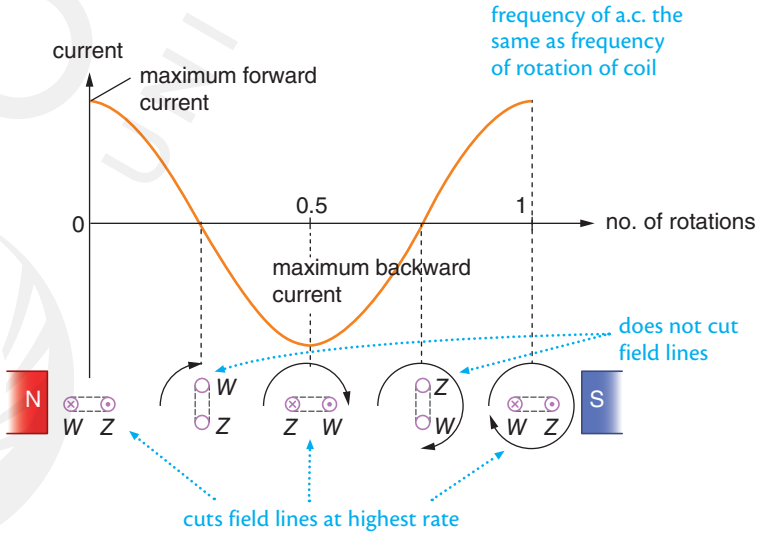



*Fig 7.2f* Current produced by an a.c. generator.

a.c. generator 交流發電機　　slip rings 匯電環

The following summarizes how simple d.c. and a.c. generators work.

**In a simple generator, when the coil is spun, the two sides of the coil cut the magnetic field lines. This induces an e.m.f. in the coil and drives a current. The current flowing in the coil reverses its direction every half turn.**

- **In a d.c. generator, the commutator reverses the connections of the coil with the external circuit every half turn. Hence, a d.c. flows in the external circuit.**
- **In an a.c. generator, the current flows through the slip rings to the external circuit. Hence, an a.c flows in the external circuit.**

## Quick check 4

→ E-version in OUP Exercise Platform is also available.

Google Forms | Microsoft Forms

**1** *True or false:*

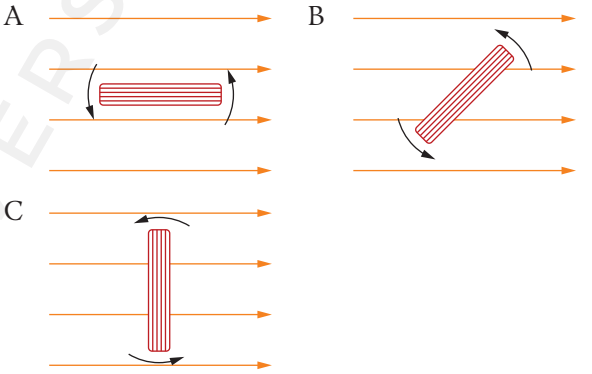
(a) In a simple a.c. generator, the coil is connected to the external circuit via a commutator. (T/F)

(b) In a simple d.c. generator, the induced current in the coil keeps flowing in the same direction. (T/F)

(c) In a simple a.c. generator, the e.m.f. induced in the coil is constant in magnitude. (T/F)

(d) In a simple d.c. generator, the frequency of the induced current is the same as the frequency of rotation of the coil. (T/F)

**2** A coil is rotating steadily in a uniform magnetic field. The figures below show the end view of the coil at different instants. At which instant is the induced e.m.f. in the coil the largest?




## C Factors affecting the output of generators

In a generator, the maximum induced e.m.f. (i.e. output voltage) increases if the coil cuts through the magnetic field lines at a higher rate. This can be achieved by

**1** using a stronger magnet,

**2** using a coil with more turns,

**3** using a coil of larger area within the magnetic field,

**4** winding the coil on a soft-iron core, ▶ This strengthens the magnetic field.

**5** rotating the coil at a higher speed (i.e. higher frequency).

## Know more

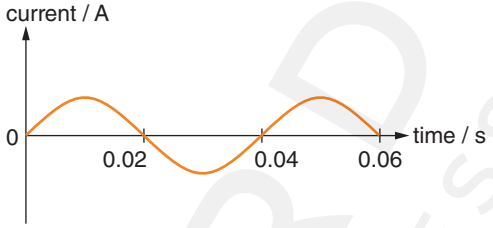
**Practical generators**

In practical generators, the coils are usually fixed and the magnets are rotated. Do you know why?

▸ p.320

## Example 6 Current produced by an a.c. generator

The figure shows how the current produced by a simple a.c. generator varies with time.




(a) State an instant when the plane of the coil becomes (i) parallel to and (ii) perpendicular to the magnetic field.

(b) What is the frequency of the current?

(c) The coil is then rotated twice as fast. Sketch the graph of current against time.

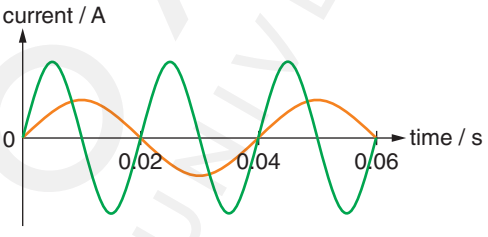
### Solution

(a) (i) 0.01 s (or 0.03 s, 0.05 s, etc.)

(ii) 0 (or 0.02 s, 0.04 s, etc.)

(b) $f = \frac{1}{T} = \frac{1}{0.04} = 25$ Hz

(c)




**Tip**

When the coil is rotated twice as fast, $f$ doubles and $T$ halves.

The maximum current produced increases since the coil cuts the magnetic field lines at a higher rate.

## Quick check 5

Google Forms Microsoft Forms

**1** Which of the following changes can increase the induced e.m.f. in a generator? Put a tick in the corresponding box(es).

- ☐ Use a coil of more turns.
- ☐ Use a coil of a larger area.
- ☐ Use a coil with a lower resistance.
- ☐ Use a stronger magnet.
- ☐ Rotate the coil faster.
- ☐ Add a soft iron core to the coil.

→ E-version in OUP Exercise Platform is also available.

## Learn by practice 1

**1** A simple d.c. generator is connected to a light bulb. The frequency of the current $I$ through the bulb is 50 Hz.

(a) Sketch a graph of $I$ against time $t$.

(b) Sketch the new $I$–$t$ graphs if

(i) the number of turns of the coil is increased;

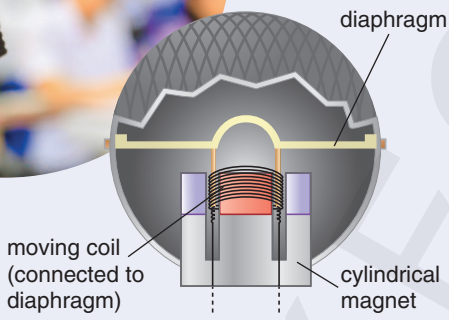
(ii) the rotating frequency of the coil is halved.

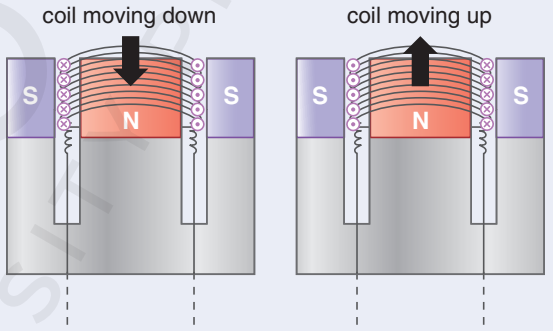
# More applications of electromagnetic induction



## Moving-coil microphone

When someone speaks or sings into a microphone, the vibration of the air causes the diaphragm to vibrate. This causes the coil to move up and down between the poles of the magnet, and induces an a.c. in the coil. The induced a.c. varies according to the frequency and the amplitude of the sound waves.



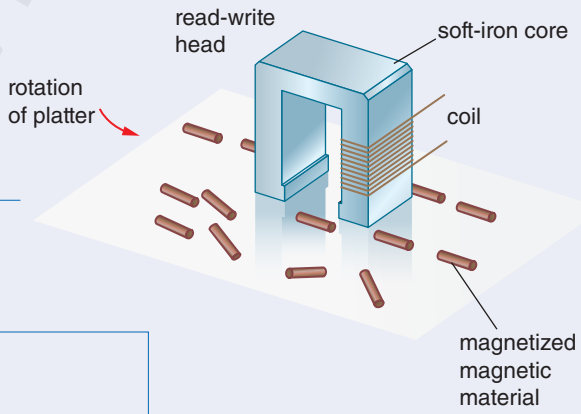




## Hard disks

A hard disk consists of several disks called platters.

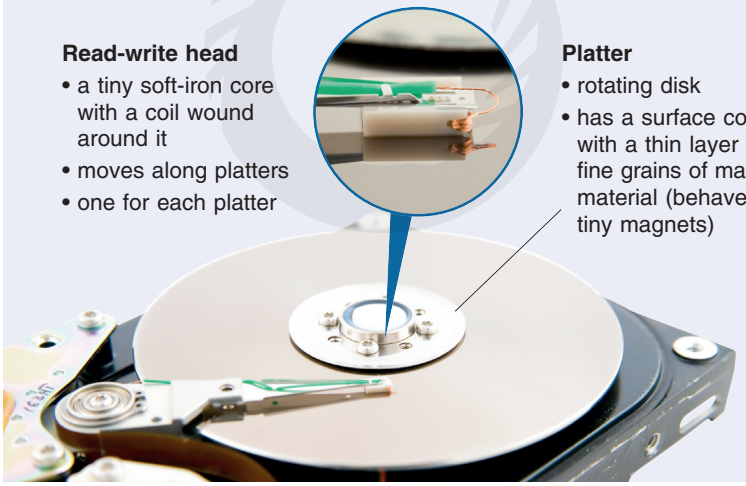
**To write**: Information is first converted into a.c. signals and sent to the coil in the read-write head. The head magnetizes and rearranges the grains of magnetic material on platters, forming a magnetic 'copy' of the information.

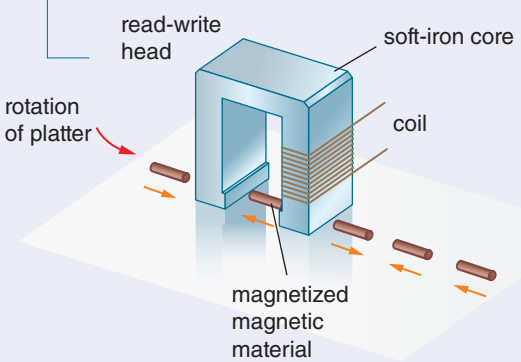
**To read**: When the head passes over the platters, the magnetic field of the grains induces a.c. in the coil of the head. The a.c. signals are then converted back to information.




**Read-write head**
- a tiny soft-iron core with a coil wound around it
- moves along platters
- one for each platter

**Platter**
- rotating disk
- has a surface coated with a thin layer of fine grains of magnetic material (behave like tiny magnets)

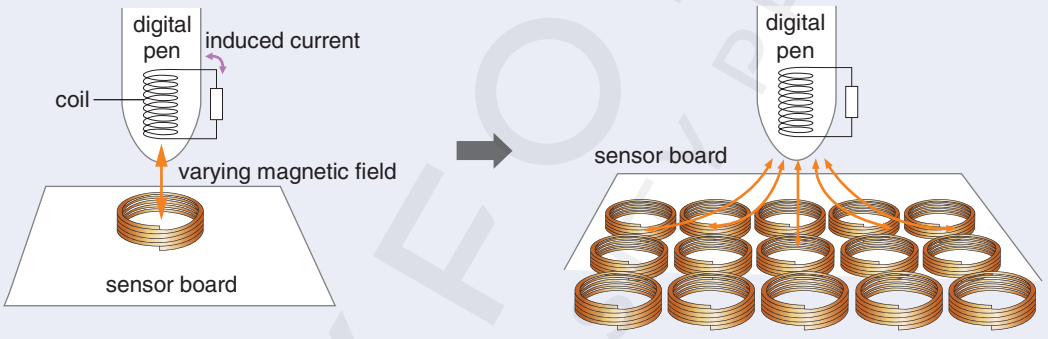






## Digital pen

One type of digital pen works with a smartphone which has arrays of coils in a sensor board. These coils (as a transmitter) keep generating a varying magnetic field. As a digital pen is moved to a position on the screen, a weak current is induced in the pen's coil. The magnetic field produced by this weak current is detected by the coils (as a receiver) in the smartphone. The pen is then located based on the magnitudes of signals detected by coils at different positions of the phone.




## The Octopus card

Each Octopus card's chip has its own coil pattern. The card machine sends a special varying magnetic field, and induces a current in the coil. The card and the card machine can then interchange data by detecting each other's magnetic field.






## Mobile payment with NFC

One of the methods to do contactless payments with smartphones is using a technology called near-field communication (NFC). It is similar to the technology used in the Octopus cards. To make a payment, a customer needs to tap the smartphone with NFC function to a payment device. Electromagnetic inductions will then occur between the smartphone and the device and information is exchanged to complete the payment.

**DSE goal**

Discuss the occurrence and practical uses of eddy currents.

**DSE exam**

14(1B)Q9, 17(1A)Q27, 18(1A)Q29, 19(1A)Q28, (1B)Q9(c), 20(1A)Q28, 21(1B)Q7, 23(1A)Q27

**Video**

→ This video demonstrates the Hands-on activity.

# 2 Eddy currents

## a Induced current inside a conductor

### Falling magnets

**Hands-on activity**

**What you need**

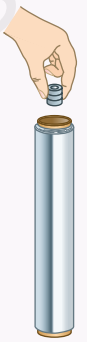
a roll of aluminium foil

neodymium magnets



**What to do**

Hold the magnets above the roll of aluminium foil and then release the magnets.



**What you find**

Does the magnet fall freely?

In this activity, the magnet falls slowly. This can be explained by electromagnetic induction. Suppose the magnet falls with the N-pole facing downwards (Fig 7.2g). Currents are induced in the roll of foil, which is a single piece of conductor, to oppose the fall of the magnet. The magnet is acted on by upward magnetic forces, so it falls slowly.

**Caution**

The currents above and below the magnet flow in opposite directions. This only happens in a large piece of conductor but not in a long coil. If the magnet is falling in a long coil and is halfway through, the e.m.f. induced above and below the magnet will cancel each other out and no current will flow.

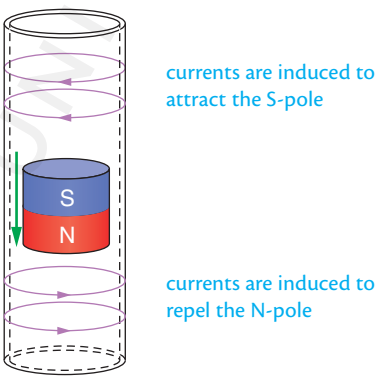



***Fig 7.2g*** *Currents are induced in the roll of aluminium foil when the magnet falls.*

The currents flow in closed loops within the conductor. They are called **eddy currents**.

**When a conductor experiences a change in magnetic field, eddy currents are induced to flow in closed loops inside the conductor to oppose this change.**

eddy current 渦電流

Let us study eddy currents more carefully in Experiment 7c.

Video

→ This video demonstrates Expt 7c.

Do not touch the ring when the a.c. power supply is switched on as it is very hot.

## Experiment 7c / Eddy current

### Part I

1 Put a coil of 1100 turns around the bottom of a retort stand. Insert an aluminium ring on top of the coil (Fig a). Connect the coil to a d.c. power supply.

2 Switch on the power supply. Observe what happens to the ring.

3 Repeat step 2 with an a.c. power supply.

4 Repeat steps 2–3 with a ring with a slit (Fig b).

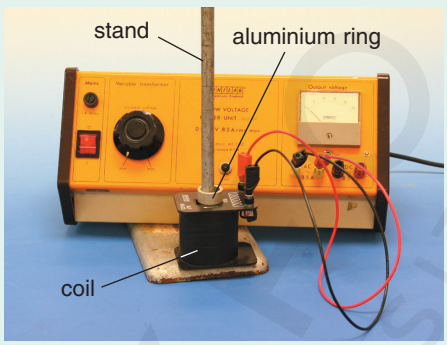



*Fig a*

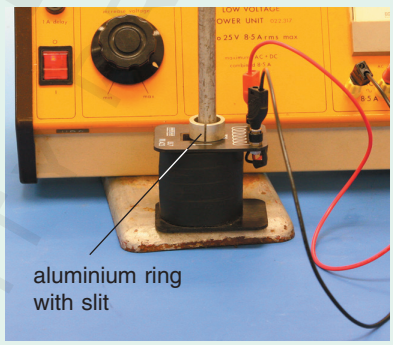



*Fig b*

### Part 2

1 Suspend an aluminium plate so that it can swing about a pivot. Place the plate between two unlike magnetic poles (Fig c).

2 Displace the plate to one side and then release the plate. Observe what happens to the plate.

3 Repeat step 2 with a plate with slots (Fig d).

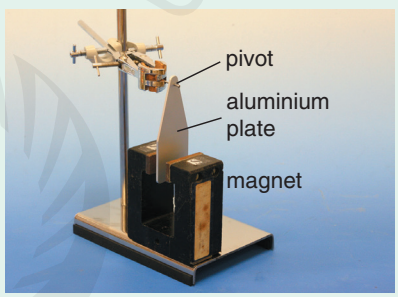



*Fig c*

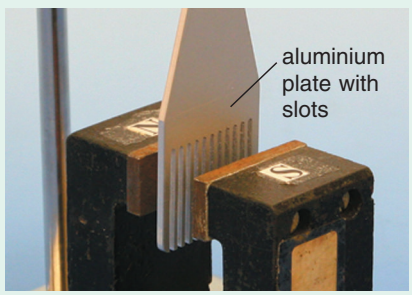



*Fig d*

### Discussion

1 What happens to the aluminium rings when using (a) a d.c. power supply and (b) an a.c. power supply?

2 What happens to the aluminium plates when they swing between a pair of slab-shaped magnets?

Unless otherwise specified, a d.c. power supply usually refers to one providing constant voltage.

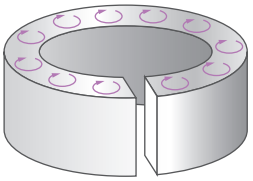
In Part 1 of Experiment 7c, when the **d.c.** power supply is **just** switched on, the magnetic field passing through the ring increases. By Lenz's law, eddy currents are induced in the ring to oppose the increasing field. The ring is repelled by the coil and so it **shoots up** (Fig 7.2h).

When the d.c. power supply remains on, the current in the coil, hence the magnetic field through the ring, become constant. As a result, the eddy currents cease to flow in the ring and the ring **drops back**.

When an **a.c.** power supply is used, the magnetic field through the ring keeps changing. Eddy currents are induced in the ring continuously to oppose the change, so the ring is kept repelled by the coil and **floats in mid-air** (Fig 7.2i).

There are eddy currents flowing in very small loops in the ring, but the magnetic forces due to these currents are very small.

When the ring with a slit is used, the slit prevents eddy currents from flowing around the whole ring, so the ring remains at rest on the coil.



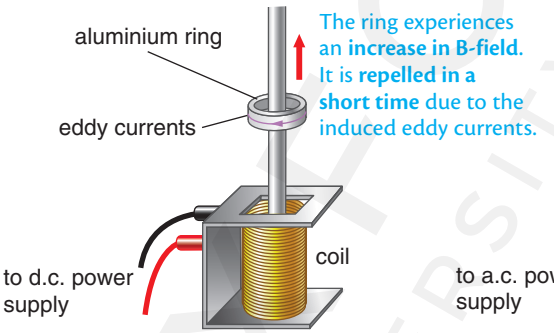



***Fig 7.2h*** *The ring shoots up when a d.c. power supply is just switched on.*

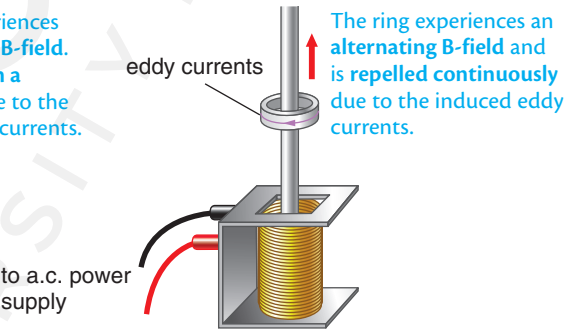



***Fig 7.2i*** *The ring keeps floating in mid-air when an a.c. power supply is switched on.*

**Video**

→ This video demonstrates how a piece of aluminium foil is attracted by a moving magnet.

In Part 2 of Experiment 7c, when the plate is approaching or leaving the magnets, it experiences a change in a magnetic field. Eddy currents are induced in the plate to oppose the change (Fig 7.2j), making the plate stop quickly.

When the plate with slots is used, the eddy currents can only flow in smaller loops. The opposing magnetic force becomes much smaller, so the plate takes a longer time to stop.

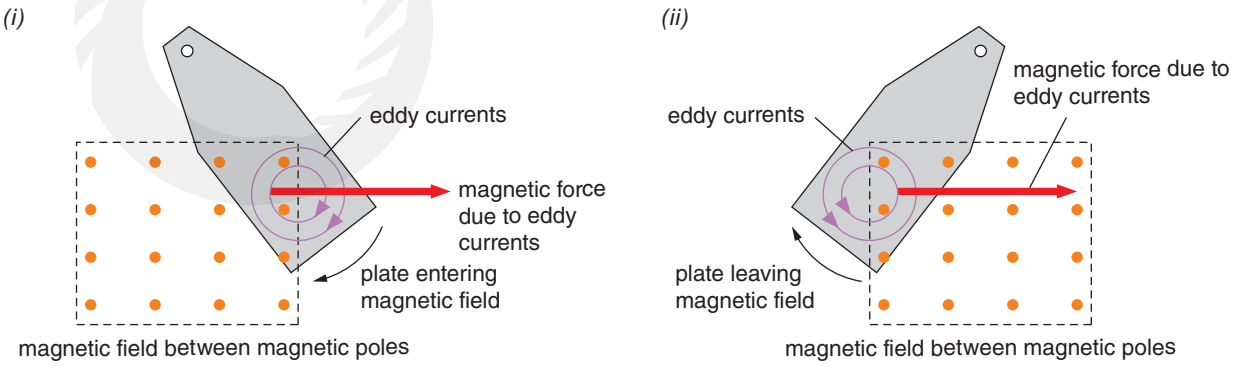


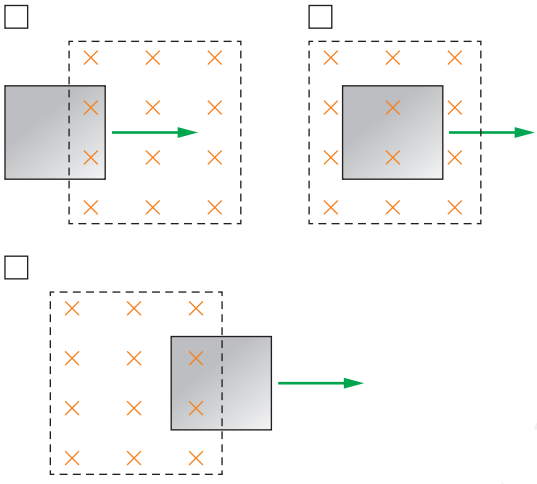

***Fig 7.2j*** *The aluminium plate (i) entering and (ii) leaving the magnetic field.*

## Quick check 6

Google Forms Microsoft Forms

1 *True or false:* Eddy currents always flow in loops. (T/F)

2 A metal plate moves to the right and passes through a uniform magnetic field. At which of the following instants will eddy currents be induced in the plate? Put a tick in the corresponding box(es).



→ E-version in OUP Exercise Platform is also available.

## Learn by practice 2

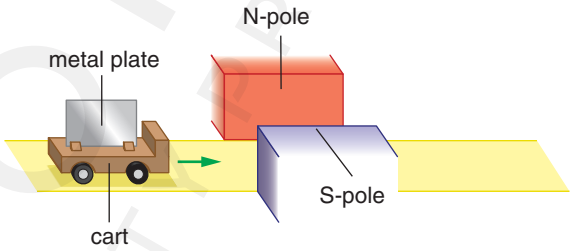
1 A metal plate is inserted vertically on a cart. The cart moves on a bench and passes between two magnets.

(a) Determine the direction of the magnetic force acting on the metal plate when the cart is moving

(i) towards the magnets,

(ii) away from the magnets.

(b) How do the answers change if the poles of the magnets are reversed?




# b Application of eddy currents

## i Induction cookers

**Cross-link**
Induction cookers will be further discussed in Book E3.

An induction cooker sets up a fast-changing magnetic field with a coil of wire carrying a high frequency a.c. Eddy currents are induced continuously in the metal cooking pot placed above (Fig 7.2k). The pot gets hot due to the **heating effect** of current and cooks the food.

**Video**

→ This video shows an expt on induction heating that explains the working principle of the induction cooker.



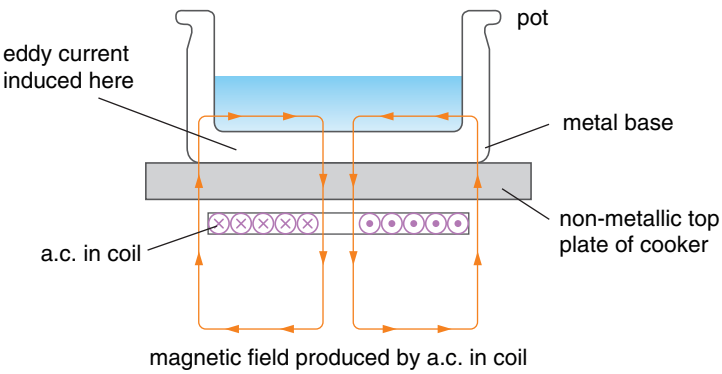



*Fig 7.2k* *An induction cooker and its working principle.*

## ii Eddy current brakes

Since there is no direct contact between the brake and the object, noise and mechanical wear can be reduced.

An eddy current brake makes use of the magnetic force due to eddy currents to slow down a fast object such as a roller coaster and a high-speed train (Fig 7.2l). It is basically a metal plate passing through a magnetic field (Fig 7.2m). The kinetic energy of the object is converted to the electrical energy of the eddy currents and dissipated finally as heat.

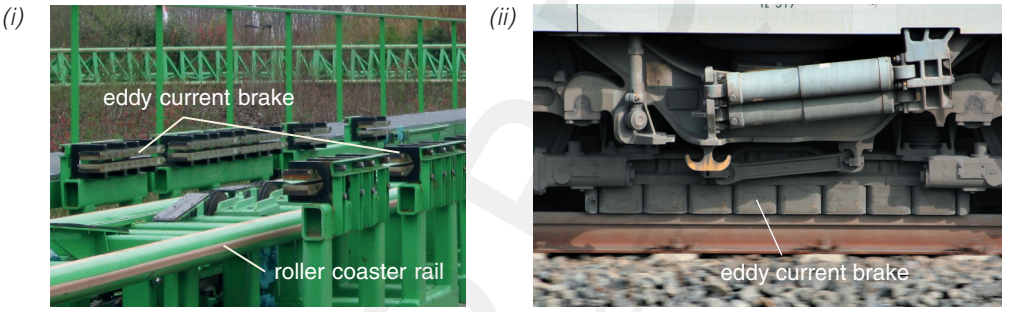



***Fig 7.2l*** *Eddy current brakes of (i) a roller coaster and (ii) a high-speed train.*

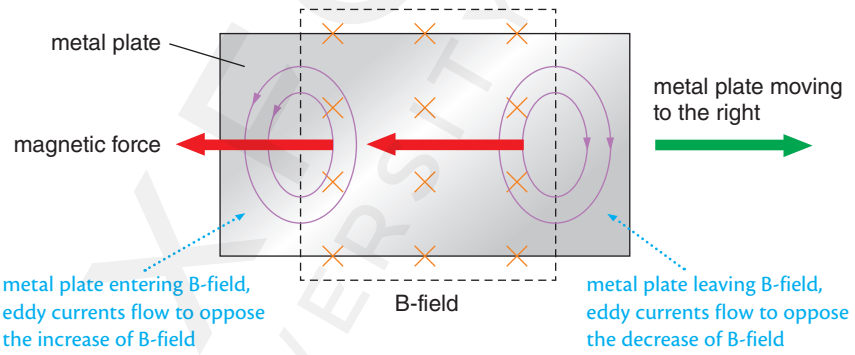



***Fig 7.2m*** *The working principle of eddy current brakes.*

May ask Ss which kind of brake can lock a vehicle in position on a slope.

The faster the object, the larger the braking force. When the object is slow, the braking force is small, so traditional frictional brake is still needed.

## iii Cap sealing

Food and medicine containers are usually sealed using eddy currents. The cap seal has a layer of aluminium foil and plastic film (Fig 7.2n). When it is placed on the container, a changing magnetic field is applied for a short time. The aluminium foil gets hot due to eddy currents and the plastic film is partly melted. When the plastic film cools down, it will be attached to the container, so the container is sealed.

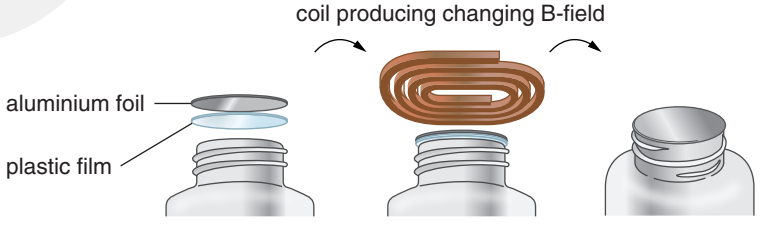
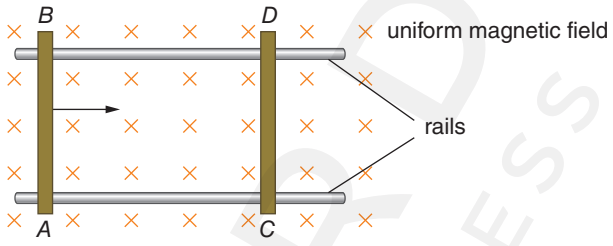



***Fig 7.2n*** *The working principle of cap sealing.*

## Writing practice

☆ HKDSE 2018 Paper 1B Q9

Metal rod *AB* is moved along two smooth horizontal conducting rails in a uniform magnetic field as shown. Metal rod *CD* is fixed on the rails.




### Question 1

Explain why an external force is needed to keep *AB* moving. (2 marks)

**Answer**

By Lenz's law, the magnetic force due to the induced current acts on *AB* to oppose *AB*'s motion. 1A

An external force is needed to balance this magnetic force to keep *AB* moving. 1A

**Or**

Work done by the external force is needed to transfer mechanical energy to electrical energy.

### Question 2

Explain whether there is current flowing through *AB* if *AB* and *CD* move at the same velocity. (2 marks)

**Answer**

No. 1A

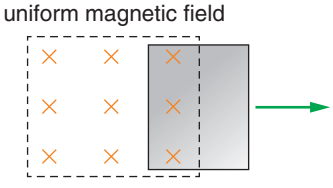
The e.m.f. induced in *AB* is the same as that in *CD*. They cancel each other out. 1A

**Or**

The amount of magnetic field lines through the loop *ABCD* does not change.

## Practise yourself

A metal plate is leaving a uniform magnetic field at a constant speed.




**1** Explain why a force is needed to pull the plate out of the field at a constant speed. (2 marks)

**2** Explain whether there is a current flow in the plate if it is completely inside the field and moves at the same speed. (2 marks)

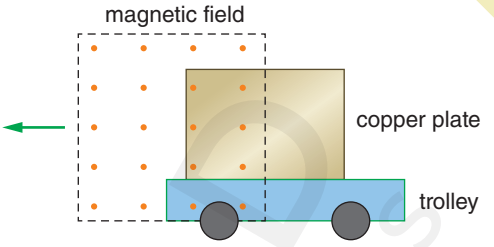
☆ HKDSE 2019 Paper 1B Q9(c)

→ This video demonstrates how this question is solved.

Video

## DSE practice 2 | Eddy currents

A copper plate is fixed on a trolley and is partly inside the uniform magnetic field between the two poles of a magnet as shown. The magnet is suddenly moved towards the left.



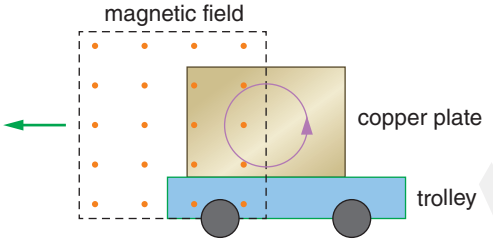

(a) Draw a circle to indicate the position where eddy currents are induced on the plate. Indicate the direction of the eddy currents with an arrow. (2 marks)

(b) Describe the motion of the trolley. (1 mark)

(c) State the energy changes when the magnet is moved. (2 marks)

### Solution

(a)




(Correct position) 1A

(Correct direction) 1A

**Common mistake** ⚠
Students may not know where the eddy currents are induced.

**Common mistake** ⚠
Students may not know that the magnetic force due to the eddy currents points to the left.

(b) It moves towards the left. 1A

(c) KE of magnet → electrical energy 1A

→ internal energy of plate + KE of trolley 1A

**Common mistake** ⚠
Students may miss the electrical energy produced in the process.

▶ Revision 7 Q31 (p.317)

## Exercise 7.2

★1

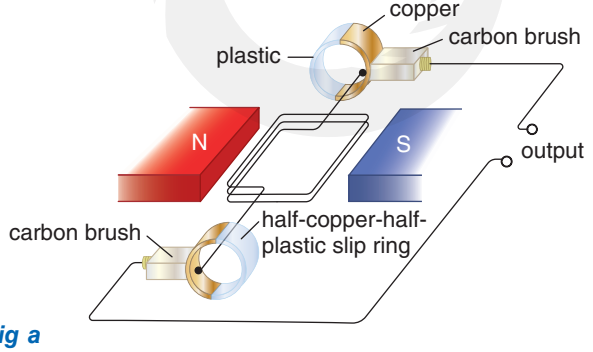
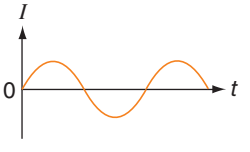
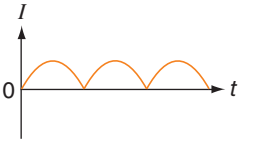
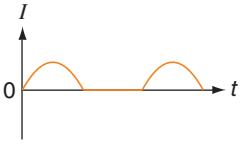


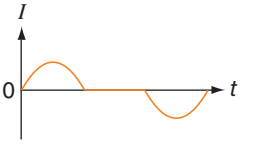

*Fig a*

Consider the generator in Figure a. If the coil is rotated uniformly, how does the output current $I$ change with time $t$?

A 


B 


C 


D 

★2

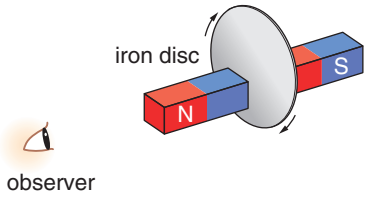
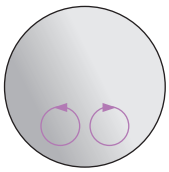


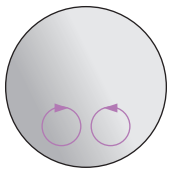

*Fig b*

An iron disc rotates clockwise at a constant speed between the poles of two bar magnets (Fig b). Which of the following diagrams best shows the eddy currents flowing in the disc from the observer's view?

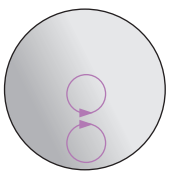
A



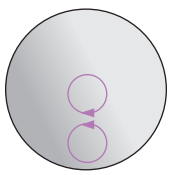
B



C



D



★3 A coil is wound on an iron rod as shown (Fig c). An aluminium ring keeps floating in mid-air on top of the coil. Which of the following types of current does the power source possibly generate?

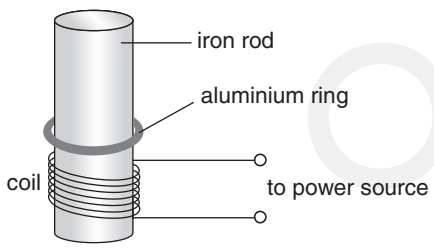



*Fig c*

(1) A.c.

(2) D.c. with steady magnitude

(3) D.c. with varying magnitude

A (1) only B (2) only

C (1) and (2) only D (1) and (3) only

★4 Figure d shows a simple generator. When it works, the coil is rotated uniformly.

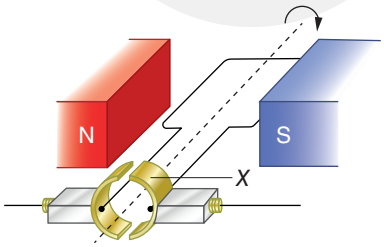



*Fig d*

(a) Name the component *X* and state its function.

(b) Sketch a graph to show how the output current varies with time over one revolution of the coil. Initially, the coil is at the position as shown in Figure d.

(c) Sketch another graph to show how the output current varies with time if the rotation speed of the coil is doubled.

★5

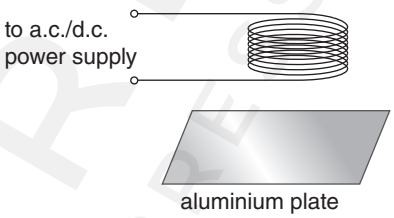



*Fig e*

A coil is held above an aluminium plate (Fig e). Will the plate become hot if

(a) a steady d.c. power supply,

(b) an a.c. power supply

is connected to the coil and is turned on for 10 minutes?

★6 Figure f shows the schematic diagram of a microphone.

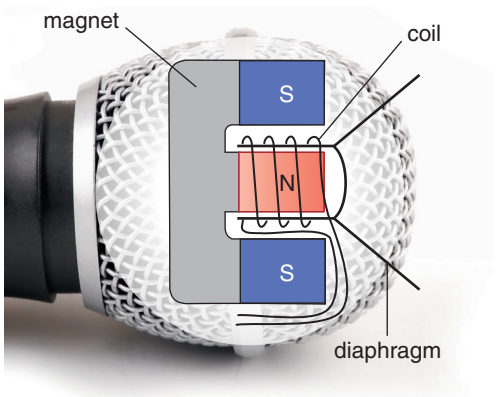



*Fig f*

(a) Describe what happens when a sound wave reaches the diaphragm.

(b) Mark in Figure f the direction of the current induced in the coil if the diaphragm is moving towards the right.

(c) State two methods to increase the amplitude of the output signal.

(d) Describe the energy conversions that take place when a sound wave enters a microphone.

Ext

# 7.3 Quantitative treatment of Faraday's law

**DSE goal**

Define $\Phi = BA\cos\theta$ and weber (Wb) as unit of magnetic flux.

Interpret magnetic field $B$ as magnetic flux density.

## 1 Magnetic flux and flux density

### a Magnetic flux

To calculate the magnitude of the induced e.m.f., Faraday introduced the term **magnetic flux**. It measures the amount of magnetic field lines crossing an area. Its symbol is $\Phi$ and its SI unit is the **weber** (**Wb**).

$\Phi$ is a Greek letter pronounced as 'phi'.

Consider a uniform magnetic field $B$ passing through a single-turn coil of area $A$ perpendicularly, i.e. $B$ is parallel to the normal of the coil (Fig 7.3a). The magnetic flux $\Phi$ through the coil is given by:

$$\Phi = BA$$

When $B$ is perpendicular to the normal of the coil (Fig 7.3b), no field line passes through the coil, so $\Phi = 0$.

We may also say that $B$ is parallel to the plane of the coil.

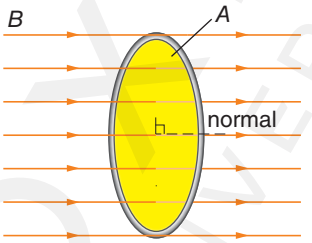



*Fig 7.3a* *B* parallel to the normal of the coil.

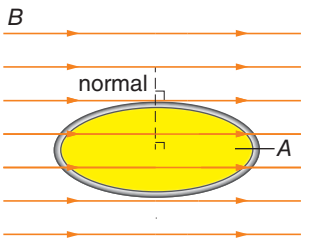
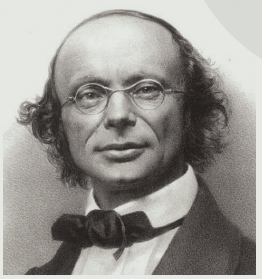



*Fig 7.3b* *B* perpendicular to the normal of the coil.

**Know more**

**Wilhelm Weber (1804–1891)**

*Wilhelm Weber* was a German physicist who set out a unified theory for electricity and magnetism into a fundamental force law in 1846. Weber linked the force between atoms to their potential energy. He also invented the *electrodynamometer*, an instrument for measuring small currents. The unit of magnetic flux is named after him.



**Know more**

**Analogy between magnetic flux and water through a ring**

You may understand magnetic flux better by thinking of a rain shower head (Fig a). The lines of water are analogous to magnetic field lines. If you put a loop under the shower head, the number of lines of water through the loop (Fig b) are analogous to the magnetic flux through a coil (Fig c).

A larger loop can enclose more lines of water. Similarly, a larger coil can enclose more field lines, hence the magnetic flux is larger.

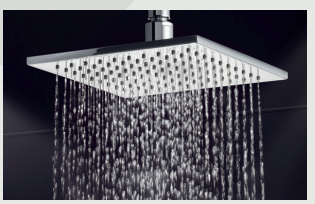


*Fig a*

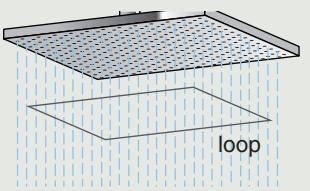



*Fig b*

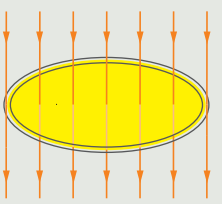


*Fig c*

magnetic flux 磁通量　weber (Wb) 韋伯　Wilhelm Weber 韋伯　electrodynamometer 力測電流計

Ext

When $B$ makes an angle $\theta$ with the normal of the coil (Fig 7.3c), only the component **parallel to the normal of the coil** contributes to the magnetic flux (Fig 7.3d). Hence, $\Phi = (B \cos \theta)A$.

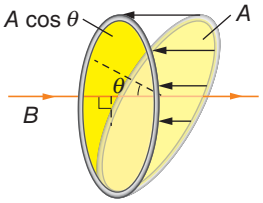
1 Wb = 1 T m$^2$ ▶

**magnetic flux** (Wb) ····· $\Phi = BA \cos \theta$ ····· **area** of coil (m$^2$)

**magnetic field** (T) ····· ····· **angle** between $B$ and normal of coil (°)

The same result can be obtained by finding the 'effective area' that is perpendicular to the B-field. In this case,

$\Phi = B(A \cos \theta) = BA \cos \theta$



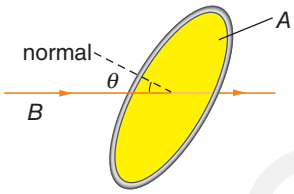




***Fig 7.3c*** *B making an angle θ with the normal of the coil.*

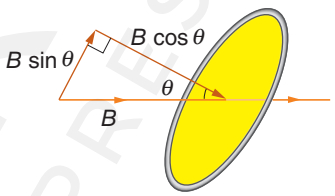



***Fig 7.3d*** *Components of B parallel and perpendicular to the normal of the coil.*

## b Magnetic flux linkage

When a uniform magnetic field $B$ passes through an $N$-turn coil, the total magnetic flux through the coil, called the **magnetic flux linkage**, is equal to $N\Phi$, where $\Phi$ is the magnetic flux through each turn (Fig 7.3e).

When the context is clear, we can simply say flux and flux linkage and drop the term 'magnetic'. ▶

**Magnetic flux linkage = $N\Phi$**

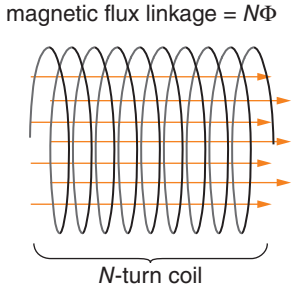



***Fig 7.3e*** *Magnetic flux linkage through an N-turn coil.*

## c Magnetic flux density

Rearranging the equation $\Phi = BA$, we obtain

$$B = \frac{\Phi}{A}$$

The strength of the magnetic field $B$ can be defined using this equation, as the magnetic flux per unit area. Hence the strength of the magnetic field is named the **magnetic flux density**. Its unit is **weber per metre squared** (**Wb m$^{-2}$**) or tesla (T).

**Cross-link**
The term magnetic flux density was introduced in Ch 5.1. ▶

**1 T = 1 Wb m$^{-2}$**

magnetic flux linkage 磁通匝鏈數　　magnetic flux density 磁通量密度

Ext

## Example 7 | Calculation of flux linkage

A 10-turn coil of area 4 cm$^2$ is placed in a uniform magnetic field of flux density $2 \times 10^{-3}$ T. Find the flux linkage through the coil if

(a) the normal of the coil is parallel to the field (Fig a);

(b) the normal of the coil makes an angle of 30° to the field (Fig b).

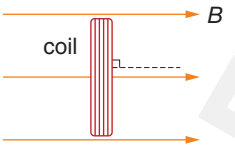



*Fig a*

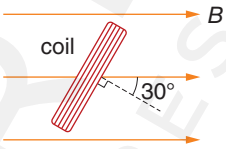



*Fig b*

### Solution

(a) Flux linkage = $N\Phi = N(BA) = (10)(2 \times 10^{-3})(4 \times 10^{-4}) = 8 \times 10^{-6}$ Wb

(b) Flux linkage = $NBA \cos\theta = (8 \times 10^{-6}) \cos 30° = 6.93 \times 10^{-6}$ Wb

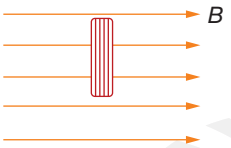
Google Forms Microsoft Forms

## Quick check 7

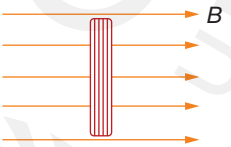
→ E-version in OUP Exercise Platform is also available.

**1** In which case is the flux through the coil larger?

A Smaller coil

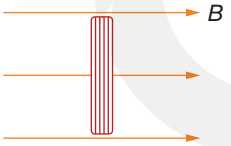



B Larger coil

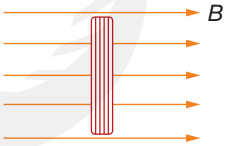



**2** In which case is the flux through the coil larger?

A Weaker B-field




B Stronger B-field



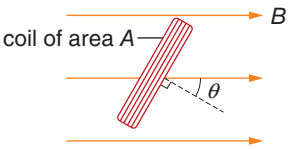

**3** Which of the following is/are the unit of magnetic flux?

☐ Wb ☐ Wb m$^{-2}$

☐ T ☐ T m$^2$

**4** *True or false:* When two single-turn coils are placed inside a uniform magnetic field, the magnetic flux through the larger coil is always larger than that through the smaller coil. (T/F)

**5** Magnetic field $B$ makes an angle $\theta$ with the normal of a coil. The coil has $N$ turns and its area is $A$.




Complete the following table.

| | $B$ / mT | $A$ / cm$^2$ | $\theta$ | $N$ | $N\Phi$ / Wb |
|---|---|---|---|---|---|
| (a) | 0.5 | 10 | 90° | 5 | |
| (b) | 2 | 16 | 60° | | $1.6 \times 10^{-5}$ |

Ext

## Learn by practice 3

**1** Lines of water fall down evenly from a rain shower head with a density of 2400 lines per $m^2$. A square loop of 10 cm × 10 cm is placed below the shower head (Fig a).

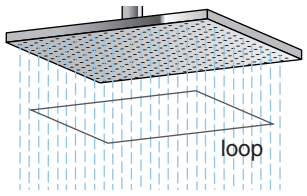



*Fig a*

Estimate the number of water lines passing through the loop when

(a) it is horizontal,

(b) it is vertical,

(c) its normal is at 60° to the vertical.

**2** A single-turn circular coil of area 10 $cm^2$ is put inside a uniform magnetic field of magnetic flux density 0.02 Wb $m^{-2}$. The magnetic field passes through the coil perpendicularly. What is the magnetic flux through the coil?

**3** A 20-turn coil of area $3 \times 10^{-4}$ $m^2$ is placed inside a uniform magnetic field of $2 \times 10^{-3}$ T. Find the flux linkage through the coil if the magnetic field

(a) is parallel to the normal of the coil (Fig b);

(b) makes an angle of 30° to the normal of the coil (Fig c).

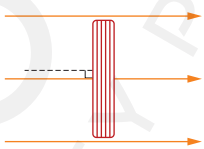



*Fig b*

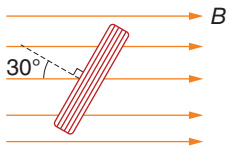



*Fig c*

**DSE goal**

State Faraday's Law as $\varepsilon = -\frac{\Delta\Phi}{\Delta t}$ and apply it to calculate average induced e.m.f.

**DSE exam**

17(1A)Q28, 19(1B)Q9(a), (b), 22(1B)Q9

# 2 Mathematical form of Faraday's law

With the concept of magnetic flux, Faraday's law can be rephrased as

**The induced e.m.f. is equal to the rate of change of magnetic flux or the rate of flux cutting.**

The induced e.m.f. can be expressed mathematically as

$$\varepsilon = -\frac{\Delta\Phi}{\Delta t}$$

induced e.m.f. (V); change in flux (Wb); change in time (s)

In the above equation, a negative sign is introduced to indicate that the induced e.m.f. opposes, or tends to oppose, the flux change causing it. Since we usually use this equation to find the magnitude of the induced e.m.f. only, we shall ignore the negative sign in calculations hereafter.

Ext

For a coil of $N$ turns, the magnitude of the induced e.m.f. becomes

$$\varepsilon = N \times \left(\frac{\Delta\Phi}{\Delta t}\right) = \frac{\Delta(N\Phi)}{\Delta t}$$

which is equal to the rate of change of the magnetic flux linkage.

## Example 8 A changing magnetic field and a rotating coil ☆ HKDSE 2019 Paper 1B Q9(b)

A 10-turn square coil of area 0.01 m$^2$ is inside a uniform magnetic field of 4 mT pointing into the page (Fig a). Find the average value of the induced e.m.f. in the coil if

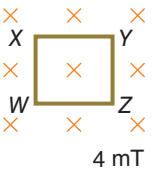



*Fig a*

(a) the magnetic field decreases to zero in 0.2 s;

(b) the magnetic field decreases to zero, reverses in direction and becomes 4 mT out of the page in 0.2 s (Fig b);

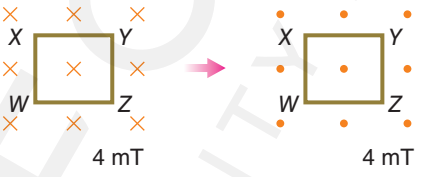



*Fig b*

(c) the coil is rotated by 180° in 0.5 s (Fig c).

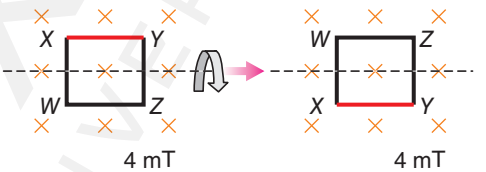



*Fig c*

### Solution

(a) $\varepsilon = \dfrac{\Delta(N\Phi)}{\Delta t} = \dfrac{\Delta(NBA)}{\Delta t} = NA \times \dfrac{\Delta B}{\Delta t} = (10)(0.01)\left(\dfrac{4 \times 10^{-3} - 0}{0.2}\right) = 2 \times 10^{-3}$ V

(b) $\Delta\Phi = \Delta(BA) = (4 \times 10^{-3})(0.01) - [-(4 \times 10^{-3})(0.01)] = 8 \times 10^{-5}$ Wb

$$\varepsilon = N\frac{\Delta\Phi}{\Delta t} = (10)\left(\frac{8 \times 10^{-5}}{0.2}\right) = 4 \times 10^{-3} \text{ V}$$

(c) $\Delta\Phi = 8 \times 10^{-5}$ Wb

$$\varepsilon = N\frac{\Delta\Phi}{\Delta t} = (10)\left(\frac{8 \times 10^{-5}}{0.5}\right) = 1.6 \times 10^{-3} \text{ V}$$

**Tip**
Use the magnitude of $\Delta B$ in calculating the magnitude of $\varepsilon$.

**Tip**
When $B$ reverses in direction, it passes through the coil in the opposite direction, so $\Phi$ changes from $BA$ to $-BA$.

**Tip**
After the coil is rotated by 180°, $B$ passes through it in the opposite direction, so $\Delta\Phi$ is the same as that in (b).

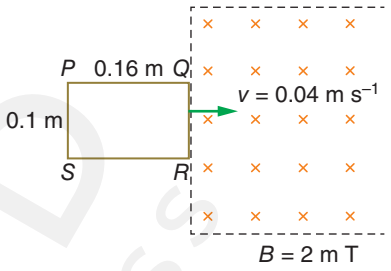
### Think about...

What is the average value of $\varepsilon$ when the coil in (c) is rotated by 90° in 0.25 s?

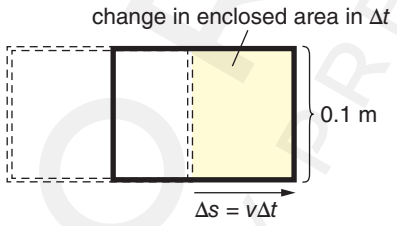
Ext

## Example 9 Rate of flux cutting

A conducting loop *PQRS* moves at a constant speed of 0.04 m $s^{-1}$ and enters a uniform magnetic field of 2 mT. Find the magnitude of the induced e.m.f. across *QR* when the coil is entering the field.




### Solution




In time $\Delta t$, the change in area of the loop inside the field is

$$\Delta A = 0.1 \times v\Delta t = 0.1(0.04)\Delta t = 0.004\Delta t$$

$$\varepsilon = \frac{\Delta\Phi}{\Delta t} = \frac{\Delta(BA)}{\Delta t} = B \times \frac{\Delta A}{\Delta t} = (2 \times 10^{-3}) \times \frac{0.004\Delta t}{\Delta t} = 8 \times 10^{-6}\ \text{V}$$

### Think about...

1 What is the direction of the induced current?

2 What is $\varepsilon$ when the loop is completely inside the field?

### Know more

**Motion of electrons**

We can get the same result by considering the motion of the free electrons in the conductor. Do you know how?

▸ p.319

We can apply the technique shown in Example 9 to find the induced e.m.f. across a conducting rod. Consider a metal rod *XY* of length *L* moving at a constant speed *v* to the right in a magnetic field *B* (Fig 7.3f).

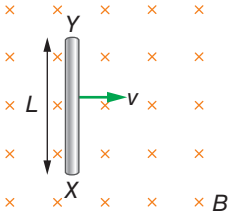



***Fig 7.3f*** *Metal rod XY moving in a B-field.*

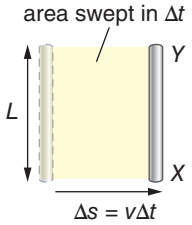



***Fig 7.3g*** *The area swept by XY in time Δt.*

In time $\Delta t$, the area swept by *XY* is $Lv\Delta t$ (Fig 7.3g). Therefore,

$$\Delta\Phi = \Delta(BA) = B\Delta A = BLv\Delta t$$

$$\varepsilon = \frac{\Delta\Phi}{\Delta t} = \frac{BLv\Delta t}{\Delta t} = BLv$$

$\varepsilon$ is proportional to *B*, *L* and *v*. This agrees with the findings of Experiment 7b on p.270. ▶

Ext

Google Forms  Microsoft Forms

## Quick check 8

→ E-version in OUP Exercise Platform is also available.

**1** A coil of $N$ turns and area $A$ is placed in a magnetic field $B$ parallel to its normal (Fig a). The change in magnetic field in time $\Delta t$ is $\Delta B$. Complete the table.

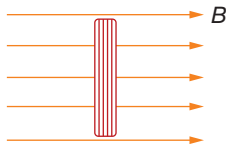



Fig a

| | $A$ / cm$^2$ | $N$ | $\Delta B$ / mT | $\Delta t$ / s | $\varepsilon$ / mV |
|---|---|---|---|---|---|
| (a) | 20 | 1 | 10 | 0.1 | |
| (b) | 5 | 200 | | 0.05 | 3 |

**2** A coil of $N$ turns and area $A$ is placed in a magnetic field $B$ (Fig b). The angle between the normal of the coil and the magnetic field changes from $\theta_1$ to $\theta_2$ in time $\Delta t$. Complete the table.

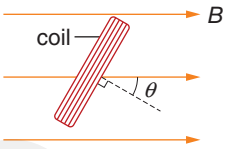



Fig b

| | $A$ / cm$^2$ | $N$ | $B$ / mT | $\theta_1$ | $\theta_2$ | $\Delta t$ / s | $\varepsilon$ / mV |
|---|---|---|---|---|---|---|---|
| (a) | 4 | 10 | 2 | 0° | 90° | 0.01 | |
| (b) | 16 | 20 | 5 | 60° | 20° | | 0.141 |

**3** A metal rod of length $L$ moves at speed $v$ and cuts a magnetic field $B$ perpendicularly. Complete the table.

| | $L$ / cm | $B$ / mT | $v$ / m s$^{-1}$ | $\varepsilon$ / mV |
|---|---|---|---|---|
| (a) | 10 | 0.5 | 8 | |
| (b) | 15 | 2 | | 1.5 |

## Learn by practice 4

**1** A coil of 5 turns and area of 0.01 m$^2$ is rotated by 40° in 1.2 s in a uniform magnetic field of 2.5 mT (Fig a). The coil is initially parallel to the magnetic field. Find the magnitude of the average induced e.m.f. in the coil.

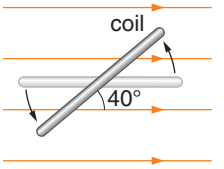



Fig a

**2** A coil of 20 turns is moving out of a uniform magnetic field of $1.6 \times 10^{-4}$ Wb m$^{-2}$ at a constant speed of 1 cm s$^{-1}$ (Fig b). Find the magnitude of the induced e.m.f. across $PQ$.

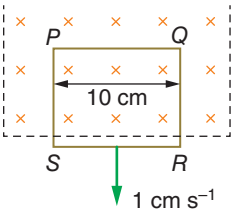



Fig b

**3** A conducting rod 10 cm long moves 8 cm to the right in 0.5 s in a uniform magnetic field of 5 mT (Fig c). What is the magnitude of the induced e.m.f. across the rod?

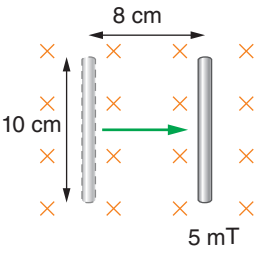



Fig c

Ext

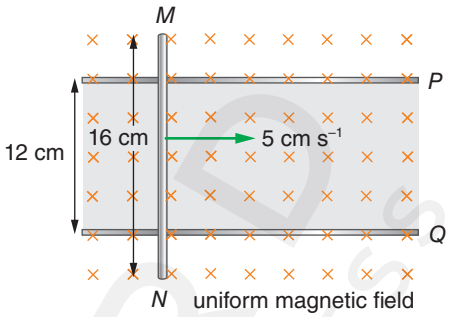
☆ HKDSE 2017 Paper 1A Q28, 2018 Paper 1B Q9

→ This video demonstrates how this question is solved.

Video

## DSE practice 3 | Metal rod moving along parallel metal rails

A metal rod $MN$ moves to the right along two smooth parallel metal rails at a constant speed of 5 cm $s^{-1}$ in a uniform magnetic field. The field is pointing into the page and of flux density 1.5 mT. The rod is 16 cm long and the two rails are 12 cm apart.




(a) Find the induced e.m.f. across $MN$. (2 marks)

(b) Does any induced current flow steadily in the rod? Explain your answer. (2 marks)

(c) A 2-Ω resistor is connected across ends $P$ and $Q$ of the rails. Assume the resistance of the other conducting parts of the set-up is negligible.

(i) Find the current through the resistor. (3 marks)

(ii) State the direction of the current in the rod. (1 mark)

(iii) After the resistor is connected, an external force is needed to keep the rod moving at a constant speed. Explain why and state the direction of the external force required. (3 marks)

### Solution

(a) $\varepsilon = BLv$

$= (1.5 \times 10^{-3})(0.16)(0.05)$ 1M

$= 1.2 \times 10^{-5}$ V 1A

**Common mistake**
Students may wrongly think that an induced current can flow steadily in an incomplete circuit, e.g. flows in a rod and stops at the end.

(b) No, 1A

because the rod is not connected to a closed circuit. 1A

(c) (i) $\varepsilon = BLv$

$= (1.5 \times 10^{-3})(0.12)(0.05)$

$= 9 \times 10^{-6}$ V 1M

**Common mistake**
Students may mistake the whole length of the metal rod in finding the induced current. As current only flows in the part of the rod between the rails, only the length between the rails should be taken.

$I = \dfrac{\varepsilon}{R}$

$= \dfrac{9 \times 10^{-6}}{2}$ 1M

$= 4.5 \times 10^{-6}$ A 1A

Use Fleming's right-hand rule.

(ii) Upwards 1A

(iii) By Lenz's law, there is a leftward magnetic force acting on the rod to oppose the rightward motion of the rod. 1A

An external force is needed to balance this magnetic force to keep the rod moving at a constant speed to the right. 1A

Therefore, the external force should point to the right. 1A

**Common mistake**
Students may mix up the magnetic force and the external force.
Note that if the circuit is open, no current will flow and no external force is needed to keep the rod moving.

▶ Revision 7 Q14 (p.312), Q30 (p.316)

Ext

**DSE goal**

Examine magnetic fields using search coil.

# 3 Search coil

Some tools were introduced in Chapter 5 for measuring the strength of magnetic field. **Search coil** is one of them (Fig 7.3h). It is used to measure a **varying** magnetic field. It is simply a small circular flat coil. It is usually connected to a CRO when in use.

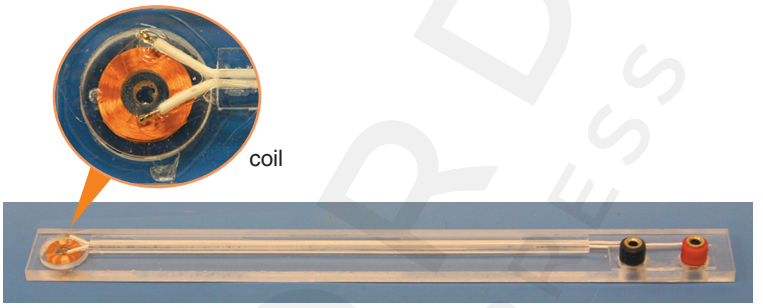



*Fig 7.3h* A search coil.

Video

→ This video demonstrates how to use a search coil to measure magnetic field.

When it is placed inside a varying magnetic field $B$, an e.m.f. $\varepsilon$ is induced across the ends of the coil. The CRO shows the waveform of $\varepsilon$. The magnitude and frequency of the magnetic field can then be deduced from the waveform.

Note that

1 When using a search coil, the plane of the coil should be perpendicular to the magnetic field (Fig 7.3i).

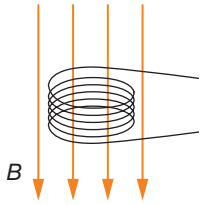



*Fig 7.3i* The coil should be perpendicular to the B-field.

2 The magnitude of the induced e.m.f. across the coil is given by

$$\varepsilon = N\frac{\Delta\Phi}{\Delta t} = NA\frac{\Delta B}{\Delta t}$$

Since $\varepsilon$ is directly proportional to $N$, increasing $N$ can increase the sensitivity of the search coil. That is why a search coil usually has thousands of turns. On the other hand, $A$ is usually small (about 1 $cm^2$) so that the magnetic field at a precise position can be measured.

3 A search coil **cannot measure a steady magnetic field** since no e.m.f. can be induced in the coil.

## Quick check 9

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

**1** A search coil connected to a CRO is used to measure a varying magnetic field. The trace shown on the CRO reaches the peak when

A the magnetic field is at maximum.

B the rate of change of magnetic field is at maximum.

C the magnetic field is at minimum.

**2** A search coil is connected to a CRO. The CRO shows a non-zero reading when the search coil is placed

A in front of a bar magnet.

B near a wire connected to an a.c. source.

C inside a solenoid connected to a steady d.c. source.

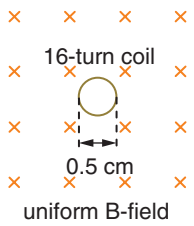
search coil 探察線圈

Ext

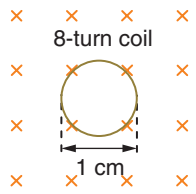
## Exercise 7.3

★**1** The following coils are placed in uniform magnetic fields of the same strength, which coil has the maximum flux linkage through it?

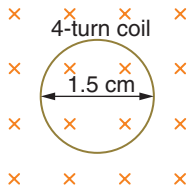
A



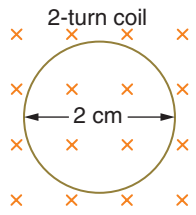

B




C




D




★**2** A search coil is placed in a varying magnetic field produced by an a.c. and an e.m.f. is induced in the search coil. Which of the following changes will increase the maximum induced e.m.f.?

(1) Increase the coil area of the search coil.

(2) Increase the number of turns in the search coil.

(3) Increase the frequency of the a.c.

A (1) only B (1) and (2) only

C (2) and (3) only D (1), (2) and (3)

★**3** A conducting ring of area $A$ rotates 90° in a uniform magnetic field so that the ring becomes parallel to the field (Fig a). The process takes 0.2 s and an average e.m.f. of $\varepsilon$ is induced in the ring. Estimate the magnetic flux through the ring before the rotation.

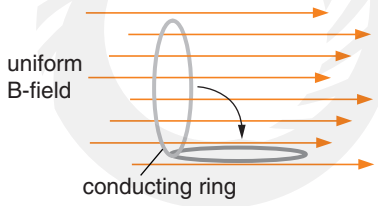



*Fig a*

A $\frac{\varepsilon}{5A}$ B $5\varepsilon A$

C $\frac{\varepsilon}{5}$ D $5\varepsilon$

★**4** A search coil connected to a CRO is used to measure the magnetic field inside a solenoid which is connected to an a.c. power supply. Figure b shows the trace on the CRO.

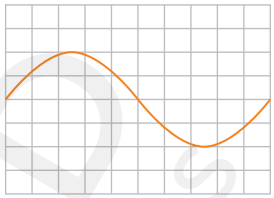
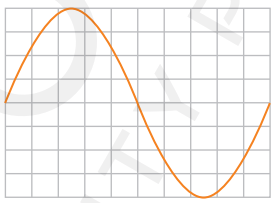


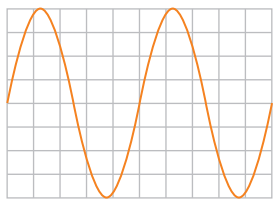
*Fig b*

If the frequency of the a.c. is doubled, which of the following figures shows the correct trace on the CRO?

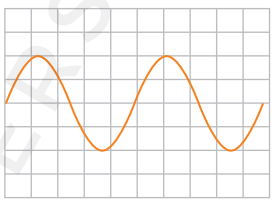
A



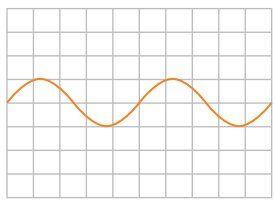
B



C



D



★**5** An aircraft is flying due east at 270 m $s^{-1}$. Its wingspan $L$ is 60 m. Suppose the Earth's magnetic field acts at 70° to the wings (horizontal) and has a strength of $1.70 \times 10^{-4}$ T (Fig c).

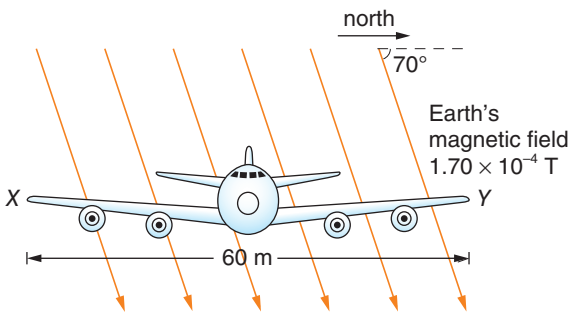



*Fig c*

(a) As the aircraft flies through the field, the tip of one wing becomes positively charged. Which tip, $X$ or $Y$, is it?

(b) Find the potential difference $V$ induced across the tips of the wings.

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Ext

*6 A rectangular coil moves at a constant speed of 5 m $s^{-1}$ and enters a uniform magnetic field of 0.02 T (Fig d). The size of the coil is 4 cm × 5 cm. The size of the magnetic field is 8 cm × 10 cm.

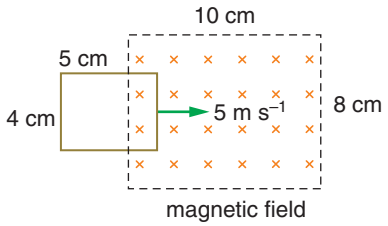



*Fig d*

(a) Find the value of the e.m.f. induced in the coil.

(b) The resistance of the coil is 5 Ω. Find the current flowing in the coil and state its direction at the instant shown.

(c) The coil is replaced by one made of the same material but with a dimension 10 cm × 5 cm (Fig e). Find the value of the e.m.f. induced in the coil.

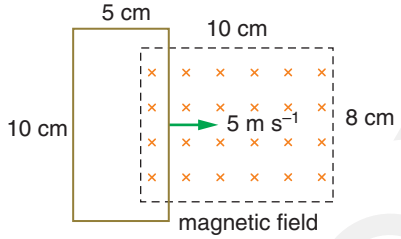



*Fig e*

*7 A metal rod is moving to the right at a constant speed of 5 m $s^{-1}$ along two smooth parallel metal rails in a magnetic field of strength 0.18 T pointing into the page (Fig f). The distance between $X$ and $Y$ is 20 cm. A light bulb of resistance 0.2 Ω is connected across the ends of the rails. Assume the resistances of the rod, the rails and the connecting wires are negligible.

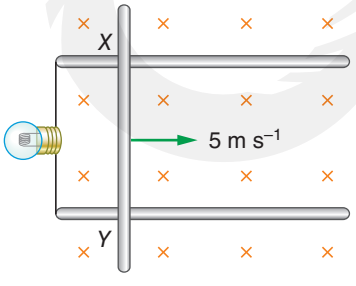



*Fig f*

(a) Find the magnitude of the induced e.m.f. across $XY$.

(b) Find the power dissipated by the bulb.

(c) Hence, find the external force needed to keep the rod moving at the constant speed.

(d) Suggest two ways to increase the brightness of the bulb.

*8 A conducting rod of length $L$ is rotating about point $O$ at a rate of 2 cycles per second in a uniform magnetic field $B$ (Fig g). Express the induced e.m.f. across the rod in terms of $B$ and $L$.

[Hint: Flux cut ∝ area swept by rod]

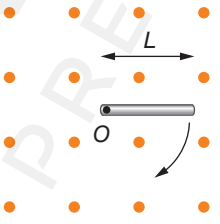



*Fig g*

*9 Figure h shows a simple d.c. generator. A uniform magnetic field of strength 0.05 T is produced between the magnetic poles. The coil has 400 turns and dimensions 2 cm × 2 cm.

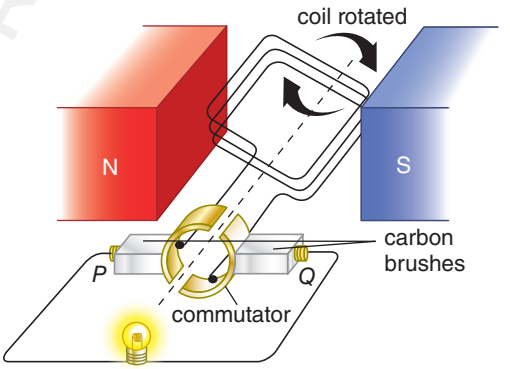



*Fig h*

(a) What is the position of the coil when

(i) the magnetic flux through the coil,

(ii) the induced e.m.f.

reaches the maximum?

(b) Indicate the direction of current through the bulb in Figure h.

(c) The coil rotates at 300 revolutions per minute. Find the average e.m.f. across $PQ$.

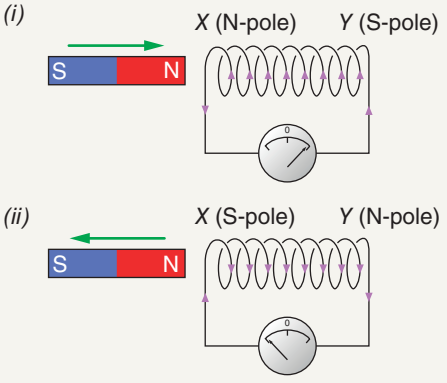
[Hint: How often does the commutator reverse the connection? What is the change in magnetic flux in this duration?]

# Key concepts 7

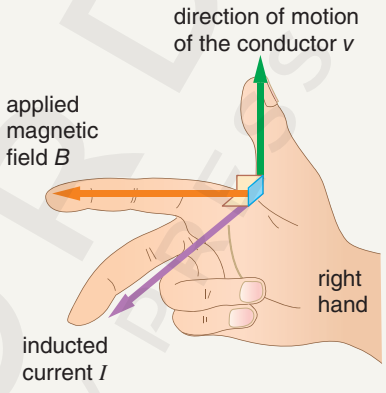
## 1 Electromagnetic induction

### Lenz's law

Induced current (or e.m.f.) opposes or tends to oppose the change causing it

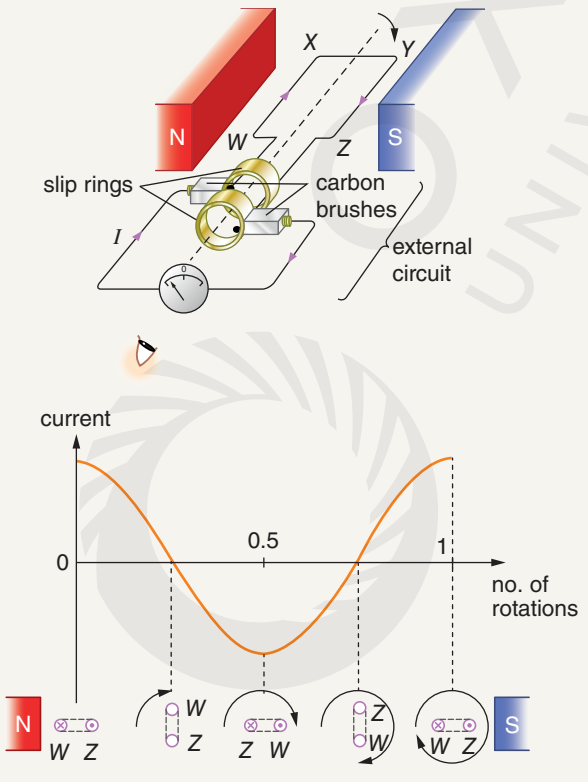



### Fleming's right hand rule

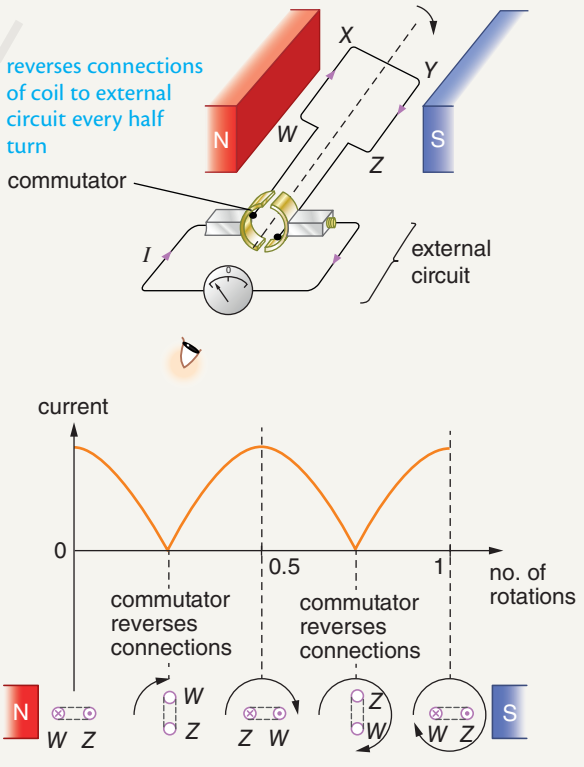



## 2 Generators

### a.c. generator



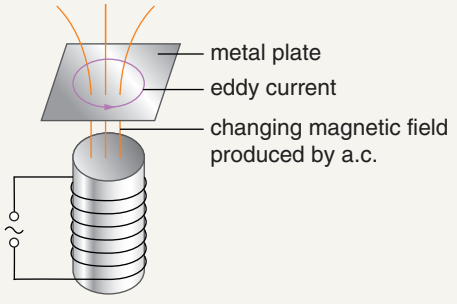

### d.c. generator




Magnitude of $I$ is maximum when coil // $B$-field;
$I$ is zero when coil $\perp$ $B$-field

## 3 Eddy currents

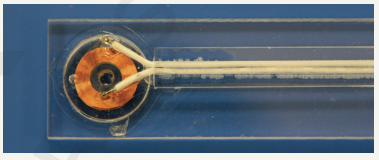
- induced currents flowing in closed loops within a conductor




- Applications: induction cookers, eddy current breaks

## 4 Search coil

- small coil with many turns
- measure varying $B$-field

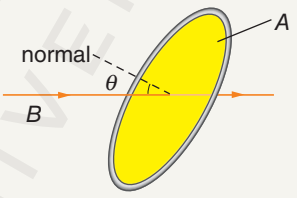




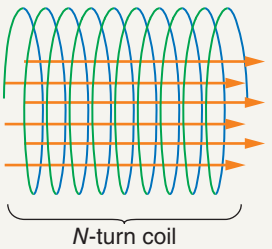

## 5 Magnetic flux Ext

**Magnetic flux $\Phi$**

- $\Phi = BA\cos\theta$
- unit: weber (Wb)

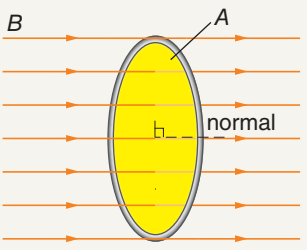



**Magnetic flux linkage** = $N\Phi$




**Magnetic flux density**

- equal to magnetic field $B$
- $B = \dfrac{\Phi}{A}$
- unit: T or Wb $m^{-2}$



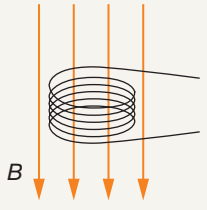
# 6 Faraday's law Ext

- For a 1-turn coil, magnitude of induced e.m.f.

$$\varepsilon = \frac{\Delta\Phi}{\Delta t}$$

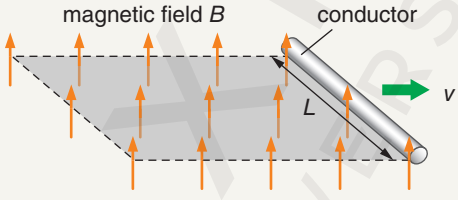
- For an $N$-turns coil, magnitude of induced e.m.f.

$$\varepsilon = N \times \left(\frac{\Delta\Phi}{\Delta t}\right) = \frac{\Delta(N\Phi)}{\Delta t}$$




- For a conductor cutting through magnetic field lines,

$$\varepsilon = BLv$$




Q29 Exam Report: In (a), very few candidates scored full marks. Many of them omitted that the ring would fall back to the coil when the current as well as the resulting magnetic field are constant and eddy currents no longer flow. It seems that some candidates did not understand Lenz's law and were not able to express their answer clearly. Some even confused the apparatus with the Lenz's law apparatus — a small magnet falling through a metal tube — and wrongly stated that the law could be demonstrated by comparing the falling speed of the aluminium ring along the retort stand under different conditions, with the switch on or off. In (b)(i), quite a number of them confused the words *flow and float*. A few candidates failed to give precise answers and stated no change or no observation etc in (b)(ii).

Q30 Exam Report: Although most candidates indicated the direction of *I* correctly in (a)(i), not many mentioned the resulting magnetic force as well as the need for an external force to balance it in order to maintain a uniform motion of the rod. In (a)(iii), may candidates employed Faraday's law to prove the equation instead of considering the mechanical power input, which was required by the question. In calculating the induced e.m.f. in (b)(ii), some candidates made mistakes in resolving the magnetic field or in converting is unit. Many did not understand the relation of induced e.m.f. and distribution of charges inside a conducting rod. (b)(iii) was poorly answered. Weaker candidates just based their answers on whether the circuit was open or closed to determine whether there would be a current. Some stated that the induced current flowing in opposite directions would cancel each other.

Q31 Exam Report: Candidates' performance was poor. In (a)(i), most candidates employed Lenz's law to explain and stated that there was a change of magnetic flux due to the decrease in magnetic field strength but forgot to mention the induced e.m.f. for driving the induced current. Some candidates were confused about 'flux change' and 'change in total flux linkage' in (a)(ii) though they managed to find the induced e.m.f. in the coil. In (b)(i), many failed to realize that the direction of magnetic flux through the coil would have been reserved after a rotation of 180°. In (b)(ii), many candidates wrongly thought that there would be no induced current at the moment when the magnetic flux through the coil was zero (while the change of magnetic flux was in fact at a maximum). Part (c) was poorly answered. Most candidates failed to indicate precisely where the eddy currents were induced. It seemed that candidates' answer to (c)(ii) were quite random – not moving, from left to right, or even rotating or oscillating.

## Key terms

1 a.c. generator 交流發電機 p.283
2 d.c. generator 直流發電機 p.281
3 eddy current 渦電流 p.288
4 electromagnetic induction 電磁感應 p.269
5 Faraday's law of electromagnetic induction 法拉第電磁感應定律 p.271
6 Fleming's right-hand rule 弗林明右手定則 p.275
7 generator/dynamo 發電機 p.281
8 induced current 感生電流 p.269
9 induced e.m.f. 感生電動勢 p.269
10 Lenz's law 楞次定律 p.272
11 magnetic flux 磁通量 p.296
12 magnetic flux density 磁通量密度 p.297
13 magnetic flux linkage 磁通匝鏈數 p.297
14 search coil 探察線圈 p.304
15 slip rings 匯電環 p.283
16 weber (Wb) 韋伯 p.296

# Revision 7

## Multiple-choice questions

★1 7.1 A conducting ring is moving away from a long straight wire carrying a stable d.c. (Fig a).

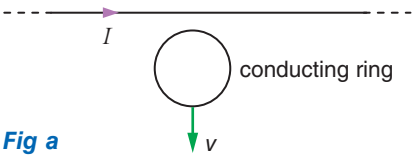



Fig a

What is the direction of the current induced in the ring?

A Clockwise

B Anticlockwise

C An a.c. is induced in the ring.

D There is no induced current.

★2 7.1 A bar magnet is being pulled away from a solenoid at constant velocity by force $F$ as shown (Fig b).

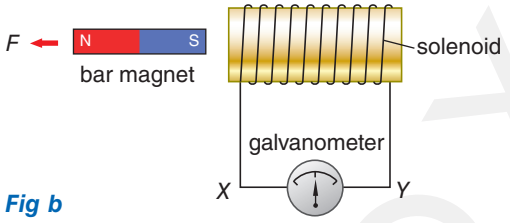



Fig b

Which of the following is/are correct?

(1) The solenoid is attracted by the magnet.

(2) An induced current flows from $X$ to $Y$ through the galvanometer.

(3) The pulling force $F$ is balanced by the magnetic force acting on the magnet.

A (3) only B (1) and (2) only

C (1) and (3) only D (1), (2) and (3)

★3 Ext 7.3 A rectangular metal frame of area $A$ is placed in a uniform magnetic field $B$. The angle between the field and the plane of the frame is 60°. The flux density of the field decreases at a constant rate from $B$ to zero in time $t$. Find the e.m.f. induced in the frame in this period of time.

A $\frac{BA}{2t}$ B $\frac{BA}{\sqrt{3}t}$

C $\frac{\sqrt{3}BA}{2t}$ D $\frac{2BA}{\sqrt{3}t}$

★4 7.1

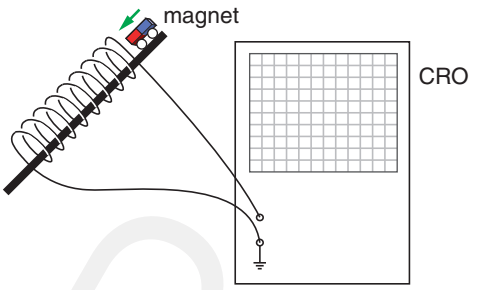
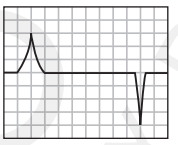
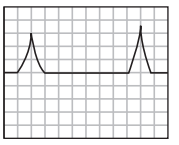
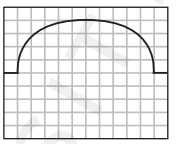


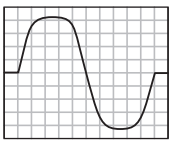

Fig c

A runway is inserted through an inclined long solenoid (Fig c) and the two ends of the solenoid are connected to a CRO. A toy car with a bar magnet installed on it runs down the runway. Which of the following diagrams best represents the trace shown on the CRO?

A 

B 

C 

D 

**(For Q5–6.)** A metal rod $XY$ of length $L$ is moving in a uniform magnetic field $B$ at a constant velocity $v$ (Fig d). The magnetic field points into the page and the rod moves to the right. The rod is perpendicular to the magnetic field and makes an angle of 30° to $v$.

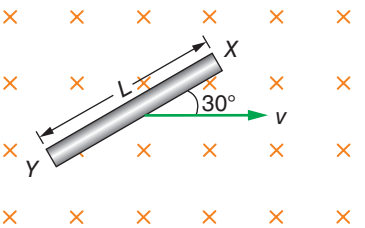



Fig d

★5 7.1 Which of the following statements is **incorrect**?

A A steady current is induced in the rod.

B There is no work done on the rod due to the magnetic field.

C End $X$ becomes positively charged while end $Y$ becomes negatively charged.

D The voltage across $XY$ will increase if $L$ increases.

★6 Ext 7.3 Find the potential difference across $XY$.

A 0 B $BLv \sin 30°$

C $BLv \cos 30°$ D $BLv$

*7 7.2 A metal plate can rotate freely around an axle. The set-up is placed between two unlike magnetic poles (Fig e). The plate is given a sharp push and starts rotating.

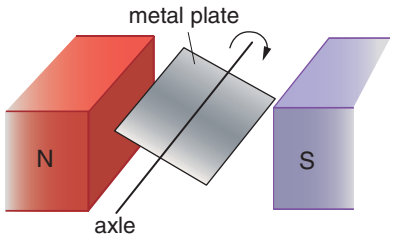


Fig e

What will happen to the metal plate afterwards?

(1) The plate will be slowed by the field.

(2) It will become hot.

(3) Eddy currents flow in the plate.

A (3) only B (1) and (2) only

C (2) and (3) only D (1), (2) and (3)

*8 Ext 7.3 Which of the following statements about a search coil is/are correct?

(1) It measures the component of magnetic field perpendicular to the plane of the coil.

(2) A search coil with more turns is more sensitive.

(3) It can be used to measure a steady magnetic field.

A (1) only B (1) and (2) only

C (2) and (3) only D (1), (2) and (3)

*9 7.1 A metal coil *PQRS* is put in a uniform magnetic field (Fig f).

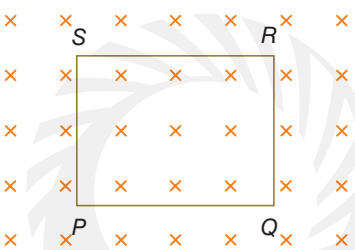


Fig f

Which of the following actions can induce a current flowing in the coil?

(1) Moving the coil towards the left within the magnetic field.

(2) Increasing the strength of the field.

(3) Rotating the coil about *SP*.

A (1) only B (2) only

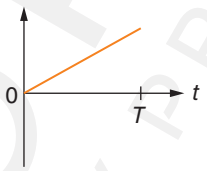
C (1) and (2) only D (2) and (3) only

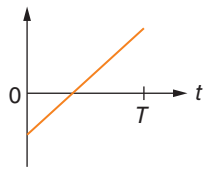
**10 7.1 A long straight wire and a conducting loop are fixed in the same plane as shown in Figure g. The current $I$ in the wire changes and the direction of the induced current in the loop is unchanged from time $t = 0$ to $t = T$.

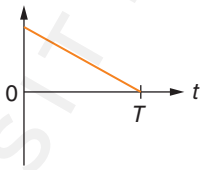


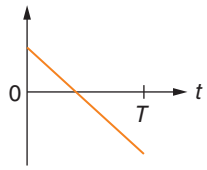
Fig g

Which of the following is/are the possible $I$–$t$ graph(s) for the current in the wire?

(1)



(2)



(3)



(4)



A (1) and (2) only B (1) and (3) only

C (3) and (4) only D (1), (2), (3), (4)

Refer p.277

**(For Q11–12.)** Figure h shows two parallel long straight wires carrying currents of the same size. A conducting loop moves to the right at a constant velocity $v$. The loop is at positions $X$ and $Y$ at time $t_1$ and $t_2$ respectively.

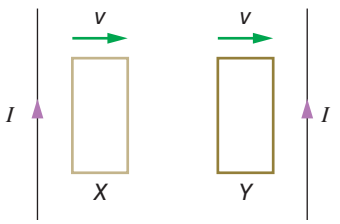


Fig h

**11 7.1 What is the direction of the induced current flowing in the loop?

| | at time $t_1$ | at time $t_2$ |
|---|---|---|
| A | clockwise | clockwise |
| B | clockwise | anti-clockwise |
| C | anti-clockwise | clockwise |
| D | anti-clockwise | anti-clockwise |

Refer p.277

**12 7.1 What is the direction of magnetic force acting on the loop?

| | at time $t_1$ | at time $t_2$ |
|---|---|---|
| A | to the left | to the left |
| B | to the left | to the right |
| C | to the right | to the left |
| D | to the right | to the right |

Refer p.277

**13 HKDSE 2015 Paper 1A Q24** 7.1

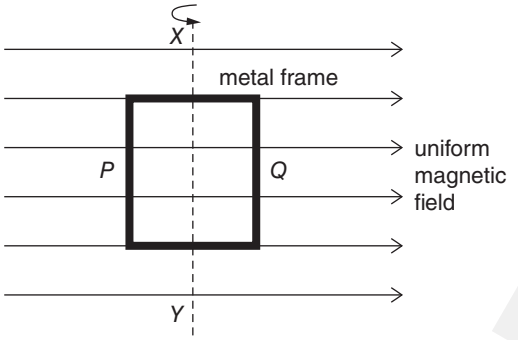



*Fig i*

A rectangular metal frame is made to rotate steadily about its axis $XY$ in a uniform magnetic field. At the instant shown, the frame is in the plane of the paper and side $P$ is moving out of the paper while side $Q$ is moving into the paper (Fig i). Which statement is **INCORRECT at this instant**?

A The induced e.m.f. in the frame is at a maximum. (17%)

B The induced current produced in the frame is flowing in anti-clockwise direction. (44%)

C The magnetic force acting on side $P$ is in a direction pointing into the paper. (26%)

D The magnetic forces acting on the frame produce a moment opposing the frame's rotation. (13%)

**14 HKDSE 2017 Paper 1A Q28** Ext 7.3

A metal rod $PQ$ of length $l$ is moving along smooth horizontal metal rails $X$ and $Y$ with constant speed $v$ in a uniform magnetic field of magnetic field strength $B$ pointing into the paper. The metal rails $X$ and $Y$ are separated by a distance of $d$ and are connected to a resistor of resistance $R$ as shown (Fig j).

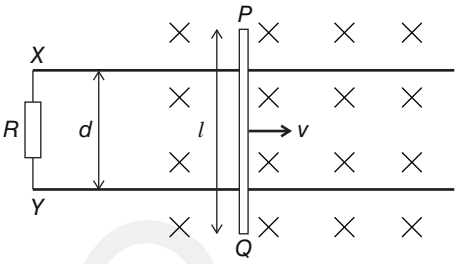



*Fig j*

Which of the following descriptions about the induced current is correct?

| | Magnitude | Direction |
|---|---|---|
| A | $\frac{Blv}{R}$ | from $X$ to $Y$ through $R$ (28%) |
| B | $\frac{Blv}{R}$ | from $Y$ to $X$ through $R$ (13%) |
| C | $\frac{Bdv}{R}$ | from $X$ to $Y$ through $R$ (46%) |
| D | $\frac{Bdv}{R}$ | from $Y$ to $X$ through $R$ (13%) |

**15 HKDSE 2020 Paper 1A Q28** 7.2

When a small strong magnet falls through a hollow aluminium tube as shown (Fig k), eddy currents are induced.

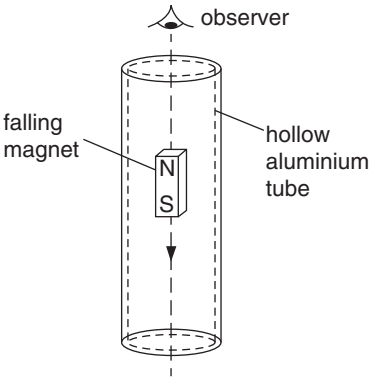



*Fig k*

Which of the following correctly describes the direction of current induced in the tube when viewed by an observer from above?

A clockwise both above and below the magnet (17%)

B anti-clockwise both above and below the magnet (19%)

C clockwise above the magnet and anticlockwise below the magnet (32%)

D anti-clockwise above the magnet and clockwise below the magnet (32%)

Exam Report: Over 40% of the candidates wrongly held that statement (3) is correct. They did not notice that no induced current flows in the rod and therefore no opposing force results.

**16** **HKDSE 2021 Paper 1A Q26**
7.1

When a copper rod *PQ* moves with a constant velocity across a uniform magnetic field as shown, an e.m.f. is induced across the rod.

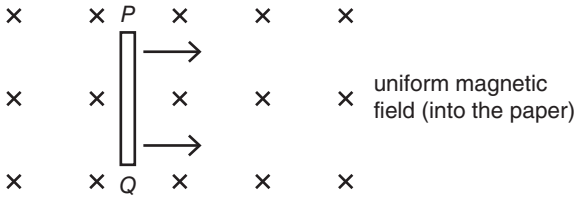



*Fig l*

Which of the following statements is/are correct?

(1) The magnitude of the induced e.m.f. depends on the length of the rod.

(2) Rod *PQ* acts like a cell providing an e.m.f. with *P* being its positive terminal.

(3) There is a force acting on the rod to opposite its motion.

A (1) only (22%) B (3) only (21%)

C (1) and (2) only (35%) D (2) and (3) only (22%)

## Conventional questions

★**17** 7.1 A magnet is suspended by an elastic string above a solenoid (Fig m). It is pulled downwards slightly and then released to make it oscillate vertically.

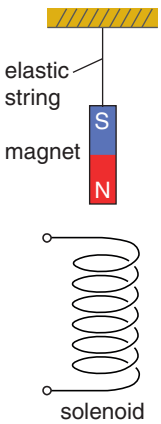



*Fig m*

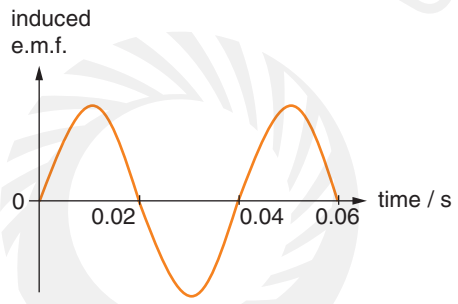



*Fig n*

(a) Figure n shows how the induced e.m.f. in the solenoid varies with time.

(i) At what times is the magnet momentarily at rest? (1 mark)

(ii) Suggest two methods to make the induced e.m.f. larger. (2 marks)

(b) What would happen if a resistor is connected across the solenoid? Explain. (4 marks)

(c) The positions of the solenoid and the magnet are interchanged, and the solenoid is made to oscillate vertically. A student claims that no e.m.f. is induced in the solenoid. Is he correct? Why? (2 marks)

★**18** 7.1 A copper ring with a slit is hung by a light string. When it is released from position 1, it swings through a square region of uniform magnetic field *B* and reaches position 2 as shown (Fig o).

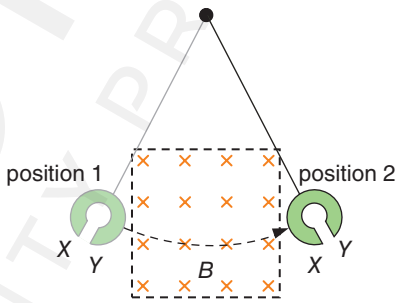



*Fig o*

(a) Which end, *X* or *Y*, is at a higher potential when the ring is

(i) entering the field, (1 mark)

(ii) leaving the field? (1 mark)

(b) The same process is repeated using a copper ring of the same size without any slit. Explain how this change affects the height of position 2. (3 marks)

★**19** 7.1 Bar magnet *X* is released to drop through a copper tube as shown (Fig p).

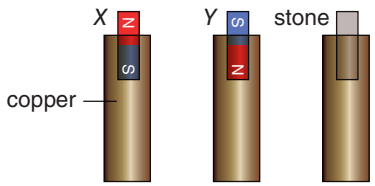



*Fig p*

(a) Draw a free-body diagram for *X*. (2 marks)

(b) If an identical bar magnet *Y* is released from the same position with its N-pole facing down, what is direction of magnetic force acting on it? Would it hit the ground at a higher speed? (2 marks)

(c) If a stone with the same size and mass is dropped from the same position, would it hit the ground at a higher speed? (1 mark)

***20** Ext 7.3 A rectangular coil *WXYZ* is moving at a constant speed of 8 m $s^{-1}$ out of a uniform magnetic field of 0.2 T (Fig q). The resistance of the coil is 0.5 Ω.

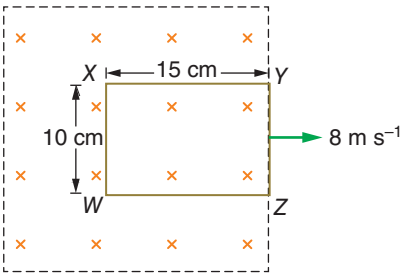



*Fig q*

(a) (i) Find the magnitude of the induced e.m.f. across *WX*. (2 marks)

(ii) Find the electrical power dissipated in the coil. (2 marks)

(b) An external force is needed to keep the coil moving at a constant velocity.

(i) Find the induced current through *WX*. (2 marks)

(ii) Find the power due to the external force. (1 mark)

***21** 7.2 A simple generator is shown below (Fig r).

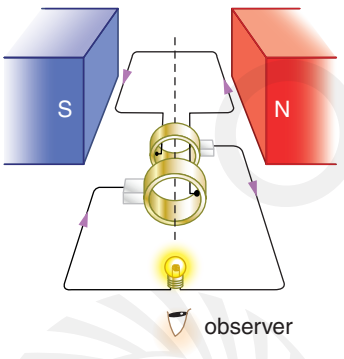



*Fig r*

(a) Is this an a.c. or a d.c. generator? (1 mark)

(b) The arrows in Figure r show the direction of current at the instant shown. In which direction is the coil rotating as viewed by the observer? (1 mark)

(c) Sketch a graph to show how the induced current varies with time over one revolution of the coil. Take $t = 0$ to be the instant shown in Figure r. (3 marks)

(d) Suggest two changes that would produce a larger current. (2 marks)

***22** Misc

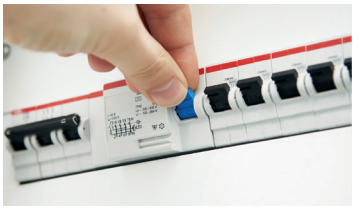


*Fig s*

A residual current circuit breaker (RCCB) (the one with blue switch in Fig s) is a device installed in the consumer unit. It would break the circuit when the currents in the live and neutral wire are different in size.

Figure t shows schematically the structure of an RCCB. The live (L) and neutral (N) wires are very close to each other. If an e.m.f. is induced in the detector coil, device *X* will break the circuit.

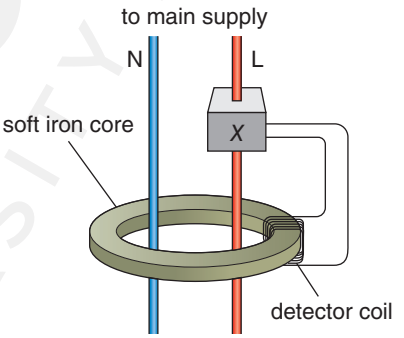



*Fig t*

(a) Why does the RCCB not break the circuit when the circuit works normally? (2 marks)

(b) Why is *X* is installed on the live wire instead of the neutral wire? (1 mark)

(c) Does a RCCB work in a steady d.c. circuit? Explain. (2 marks)

***23** Misc Figure u shows the structure of a simple d.c. generator. The coil has *N* turns and its area is *A*.

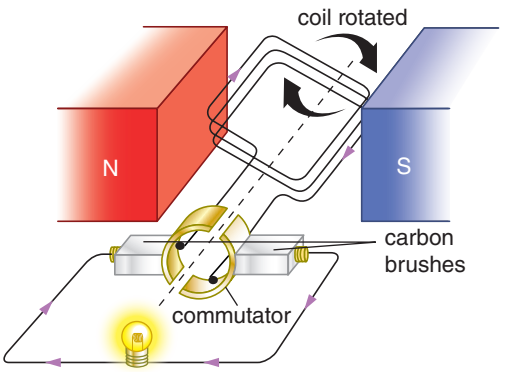



*Fig u*

(a) Explain why the size of the current induced in the coil varies from time to time when the coil is rotated steadily. (1 mark)

(b) Figure v shows how the magnetic flux linkage through the coil varies with time. Ext

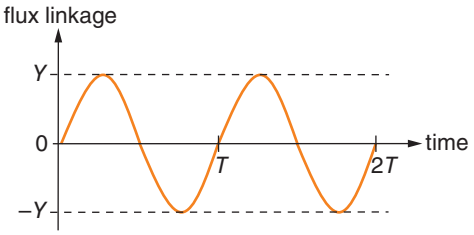



*Fig v*

(i) State the orientation of the coil when the flux linkage is at its maximum. (1 mark)

(ii) Find the magnetic flux density between the poles. (2 marks)

(iii) Sketch a graph to show how the current flowing through the light bulb varies with time from $t = 0$ to $2T$. (2 marks)

*24 7.2 Figure w shows the structure of a bicycle dynamo.

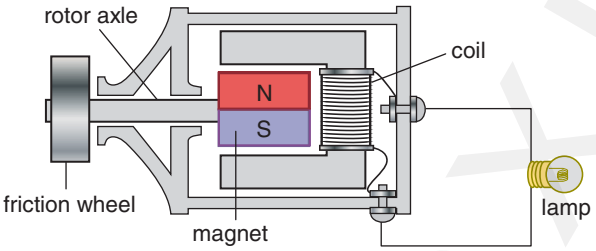



*Fig w*

(a) When the dynamo works, the magnet rotates while the coil is fixed. Give one of its advantages over a moving-coil dynamo. (1 mark)

(b) The dynamo is used to power a bicycle lamp.

(i) Describe the energy conversion that takes place when a person pedals a bicycle to light the lamp. (2 marks)

(ii) It is a bit harder to pedal the bicycle when the lamp is on. Explain why. (2 marks)

*25 Ext 7.3 In Figure x, a rectangular metal frame $PQRS$ of length 16 cm and width 10 cm is moved by an external force towards a uniform magnetic field at a constant velocity $v$. Figure y shows how the induced e.m.f. $\varepsilon$ in the frame varies with time $t$.

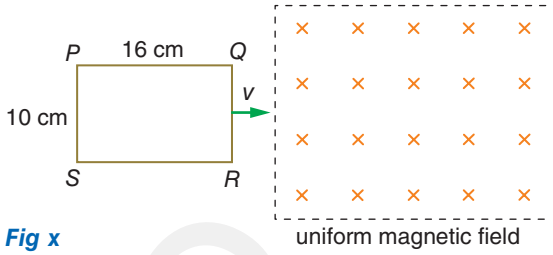



*Fig x*

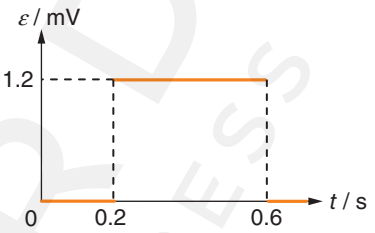



*Fig y*

(a) What is the physical meaning of the area under the $\varepsilon$–$t$ graph? (1 mark)

(b) Find the magnitude of the magnetic flux density. (2 marks)

(c) Describe the energy conversion from $t = 0.2$ s to $t = 0.6$ s. (2 marks)

*26 Misc Figure z shows a rechargeable electric toothbrush on a charging base. The base is connected to the mains supply.

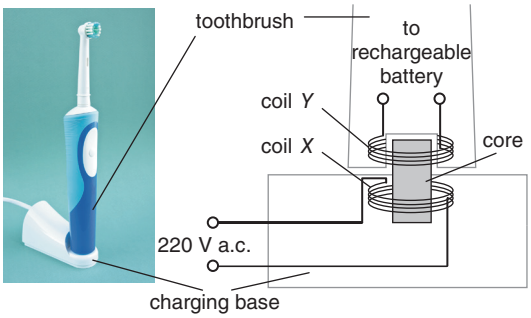



*Fig z*

(a) Describe how the battery in the toothbrush is charged. (4 marks)

(b) Figure aa shows the variation of the flux density of the magnetic field at coil $Y$ produced by coil $X$. Ext

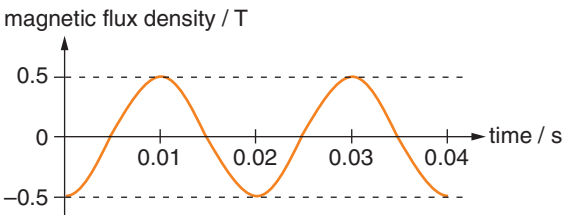



*Fig aa*

It is known that coil $Y$ has 122 turns and its radius is 1.4 cm. Estimate the average e.m.f. induced in coil $Y$. (3 marks)

**27 Ext Misc A circular wire frame of radius 15 cm is placed normally to a uniform magnetic field of 0.04 T pointing into the page (Fig ab). It is then rotated by 38° in 0.3 s about axis *X*.

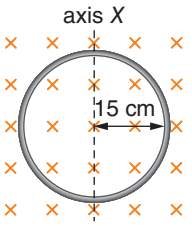



*Fig ab*

(a) Find the average e.m.f. induced in the frame within this period of time. (3 marks)

(b) The resistivity of the wire is $1.7 \times 10^{-8}$ Ω m and the diameter of the wire is 2 mm. Find the average current induced in the frame and state its direction. (3 marks)

(c) If the resistivity of the wire is higher, how will the average e.m.f. and current induced in the frame change? Explain. (2 marks)

Refer p.300

**28 Ext 7.3 In Figure ac, a square coil of dimensions 10 cm × 10 cm is moved by an external force through a uniform magnetic field of strength 25 mT. It moves at a constant speed in the whole process. It enters the field at $t = 0$ and side *PS* leaves at $t = 0.5$ s.

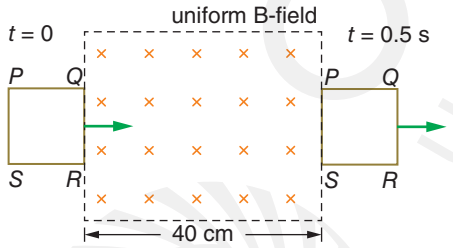



*Fig ac*

(a) What is the maximum magnetic flux through the coil within this 0.5 s? (2 marks)

(b) Sketch a graph to show how the magnetic flux through the coil varies from $t = 0$ to $t = 0.5$ s. (4 marks)

(c) State the physical meaning of the slope of the graph in (b). (1 mark)

(d) State in which durations an induced current flows within the coil. (2 marks)

(e) The total resistance of the coil is 0.004 Ω.

(i) Find the power dissipated in the coil when an induced current flows. (3 marks)

(ii) Sketch the induced current against time graph by regarding clockwise as positive. (4 marks)

(iii) Find the magnitude of the external force at $t = 0.05$ and $t = 0.2$ s. (3 marks)

Refer p.303

See p.309 for the Exam Report.

**29 HKDSE 2014 Paper 1B Q9** 7.2

(a) You are given a low voltage d.c. supply, an aluminium ring, a switch, a coil of 1100 turns and a retort stand arranged as shown. Use three connecting leads to complete the connections among the apparatus in the figure and describe how to demonstrate Lenz's law in electromagnetic induction. State and explain the observation. (6 marks)

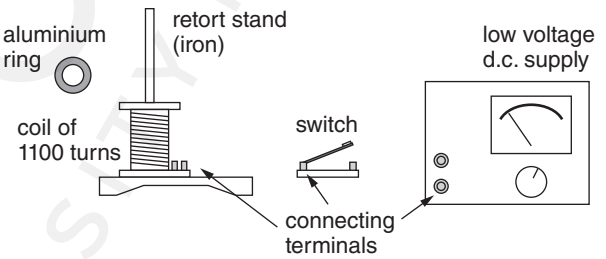



*Fig ad*

(b) Describe what would be observed if the experiment in part (a) is repeated with

(i) a low voltage a.c. supply; (1 mark)

(ii) a low voltage a.c. supply and an aluminium ring with a slit cut through it as shown [◯]. (1 mark)

See p.309 for the Exam Report.

**30 HKDSE 2018 Paper 1B Q9** Misc

(a)

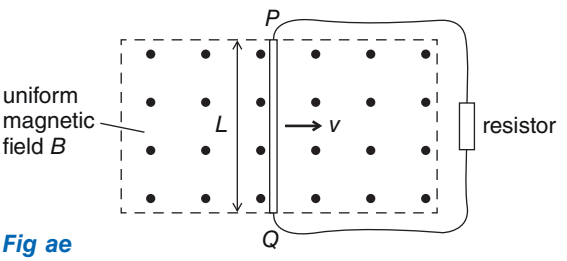



*Fig ae*

Figure ae shows a metal rod *PQ* of length *L* moving with constant velocity *v* across a uniform magnetic field *B* pointing out of the paper. An e.m.f. $\xi$ is induced across rod *PQ* as it cuts the field lines. When the rod is connected to a resistor outside the field, a current *I* flows in the circuit.

(i) Indicate the direction of *I* in Figure ae. (1 mark)

(ii) Explain why an external force $F$ is required to maintain the uniform motion of rod $PQ$. Find $F$ in terms of the physical quantities given. (3 marks)

(iii) This set-up works as a generator. By considering the mechanical power input by external force $F$ to the set-up, show $\xi = BLv$. (2 marks)

(b) At a certain place, the Earth's magnetic field runs along the S-N direction such that the field lines make an angle $\theta$ with the horizontal as shown in Figure af.

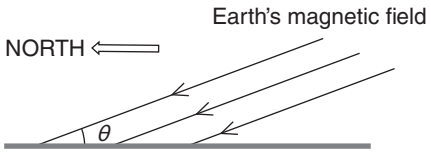



*Fig af*

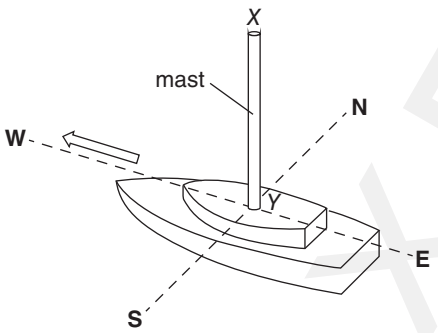



*Fig ag*

A ship with a vertical aluminium mast sails at sea along a straight course to the west as shown in Figure ag. As a result, an e.m.f. is induced across the mast $XY$.

(i) Explain why it is **only the horizontal component** of the Earth's magnetic field that is cut by the mast which gives rise to this induced e.m.f. (1 mark)

(ii) Given: length of mast $XY$ = 20 m, speed of the ship = 6 m $s^{-1}$, Earth's magnetic field = 50 μT, $\theta = 30°$

Referring to (a)(iii), calculate the e.m.f. induced across $XY$ and state whether the distribution of free electrons along the mast is more at end $X$, more at end $Y$ or uniform along $XY$. (3 marks)

(iii) Suppose $X$ and $Y$ are connected by a cable running side-by-side with the mast so that they form a complete circuit. Explain whether there will be a current passing through it. (2 marks)

See p.309 for the Exam Report.

**31** Misc

## HKDSE 2019 Paper 1B Q9

A rectangular coil $PQRS$ of 20 turns, each having an area of 0.005 $m^2$, is placed in a uniform magnetic field $B$ of strength 0.3 T pointing into the paper as shown in Figure ah.

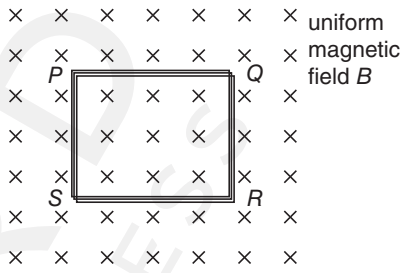



*Fig ah*

(a) The strength of the magnetic field decreases uniformly to zero within 0.5 s.

(i) Explain why a current would be induced in the coil. (2 marks)

(ii) Ext Calculate the change in total magnetic flux linkage through the coil and the value of the induced e.m.f. $\xi$ in the coil. (3 marks)

(b) Now the coil is rotated uniformly about an axis through 180° as shown in Figure ai within 0.5 s.

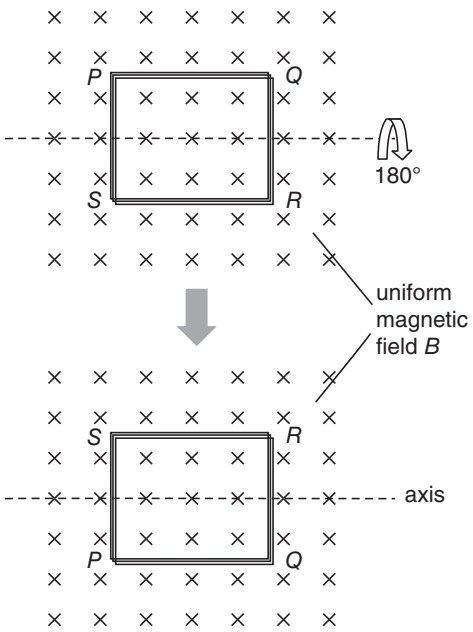



*Fig ai*

(i) Ext State the value of the change in total magnetic flux linkage through the coil in this case. (1 mark)

(ii) At the instant the coil rotated through 90°, would the induced current flow in the direction $PQRS$, $PSRQ$ or is there no induced current in the coil? (1 mark)

(c) Figure aj shows a thin rectangular aluminium plate suspended by a long string. The plate is partly inside a uniform magnetic field provided by a strong magnet.

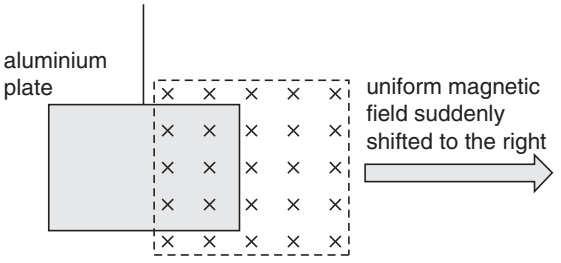



*Fig aj*

The magnet, which is not in contact with the plate, is suddenly shifted to the right.

(i) On Figure aj, draw a small circle at the location where eddy currents are induced on the aluminium plate. Use an arrow to indicate the direction of current. (2 marks)

(ii) Describe the subsequent motion of the aluminium plate, if any. (1 mark)

☆ Part (c) of the original question is omitted.

**32 HKDSE 2021 Paper 1B Q7**

7.2

Read the following passage about **eddy currents** and answer the questions that follow.

Eddy currents are induced by changing magnetic fields. They flow in closed loops in conductors like swirling eddies in a stream, perpendicular to the direction of the magnetic field. They are commonly applied in braking known as 'eddy braking'.

The heating effect of eddy currents is used in induction heating devices, such as induction cookers. The resistance felt by the eddy currents in a conductor causes Joule heating. However, for applications like motors and transformers, this heat is considered as a waste of energy and as such, eddy currents need to be minimized.

Eddy currents can be removed by cracks or slits in the conductor, which prevent the current loops from circulating. This means that eddy currents can be used in detecting defects in materials. The magnetic field produced by the eddy currents is measured, where a change in the field reveals the presence of an irregularity in the material.

(a) (i) In Figure ak, a permanent magnet with north pole facing downwards is held stationary. A metal sheet moves past the magnet (the direction of movement is not shown) and eddy currents are induced as shown. Briefly explain why eddy currents are induced and state whether the metal sheet is moving forward, backward, towards the left or towards the right. (2 marks)

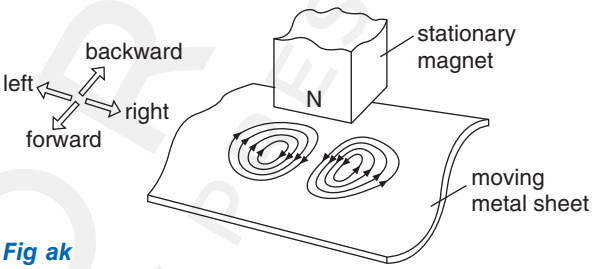



*Fig ak*

(ii) State the energy changes in the process in which the metal sheet is slowing down to stop. (2 marks)

(iii) Although eddy braking has the advantage of being contactless, traditional frictional braking cannot be totally replaced by eddy braking. Why? (1 mark)

(b) An induction cooker of rating '220 V, 2000 W' operates for 15 minutes. How much should be paid if 1 kW h of electrical energy costs $1.1? (2 marks)

(c) Eddy currents can be used to detect defects in materials. When there is a crack in a material, how would the magnetic field due to eddy currents change? Explain briefly. (2 marks)

Q32 Exam Report: This question tested candidates' knowledge and understanding of eddy currents and their applications. The performance was poor in general. In (a)(i), not many candidates identified the correct direction of movement of the metal sheet. Most knew how eddy currents were induced, however, some forgot to mention the relative motion between the plate and the magnet. In (a)(ii), quite a number of the candidates omitted the electrical energy produced in the process. In (a)(iii), very few realized that eddy braking operates only when the vehicle is moving. Some wrongly held that frictional braking was more effective or eddy braking would generate an enormous amount of thermal energy. Part (b) was well answered. Some candidates misunderstood part (c), using the production of eddy currents due to changing magnetic field instead of the magnetic field generated by eddy currents in their answer.

## Know more

### Michael Faraday (p.269)

Michael Faraday conducted many experiments on electromagnetic induction, one of which was similar to Experiment 7a on p.268. In that experiment, he moved a current-carrying coil *A* in and out of another coil *B* connected to a galvanometer (Fig a). He found that a current was measured in coil *B* as coil *A* moved.

He also built the prototype of a d.c. generator, known as Faraday disc (Fig b), in 1831. It consisted of a copper disc rotating between the poles of a horseshoe magnet. It produced only a very low d.c. voltage (between the centre of the disc and the rim) when the disc rotated.

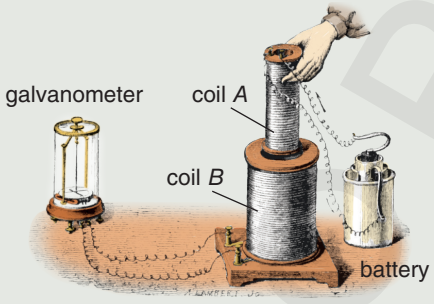



*Fig a*

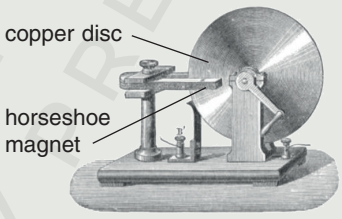



*Fig b*

### Motion of electrons (p.301)

When a conductor moves to the right at speed *v*, the free electrons inside also move to the right. By Fleming's **left**-hand rule, they are pushed downwards by magnetic forces (Fig c).

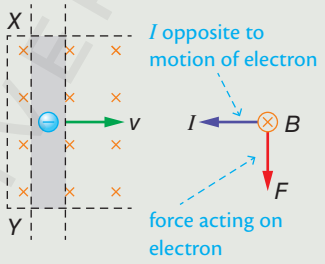



*Fig c*

Since the free electrons move downwards in the conductor, the induced current flows upwards.

If the conductor is not connected to an external circuit, electrons will accumulate at the lower end, making this end negatively charged (lower potential) and the upper end positively charged (higher potential). This sets up an electric field *E* pointing from the upper end to the lower end. If the length of the conductor is *L* and the induced e.m.f. across the conductor is $\mathcal{E}$,

This is like the electric field between two charged parallel plates. ▶

$$E = \frac{\mathcal{E}}{L}$$

Electrons stop flowing when the magnetic forces acting on them is balanced by the electric forces. Therefore,

$$Bev = Ee = \frac{\mathcal{E}}{L} \times e$$

$$\mathcal{E} = BLv$$

Note that the balance is reached in a very short time. The flow of electrons lasts for such a short time that the current in the conductor is negligible.

## Know more

### Practical generators (p.285)

A bicycle dynamo has a fixed coil and rotating permanent magnets (Fig a and b). This design avoids the circuit from making connection via a moving contact, which may produce sparks.

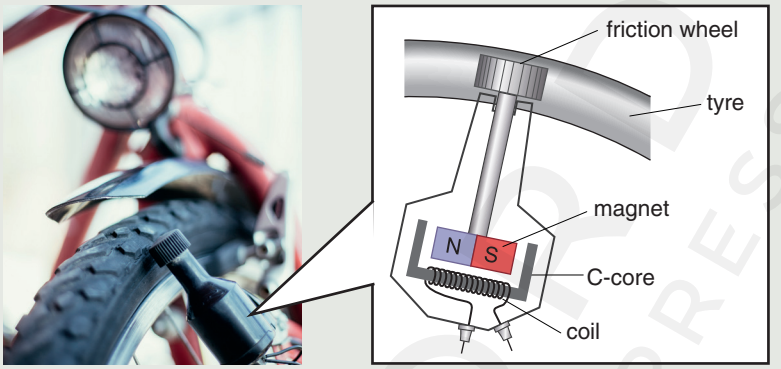



*Fig a*




*Fig b*

In a power station, the generator has a rotating electromagnet (called *rotor*) to set up a varying magnetic field and fixed sets of coil (called *stator*) around the rotor (Fig c).

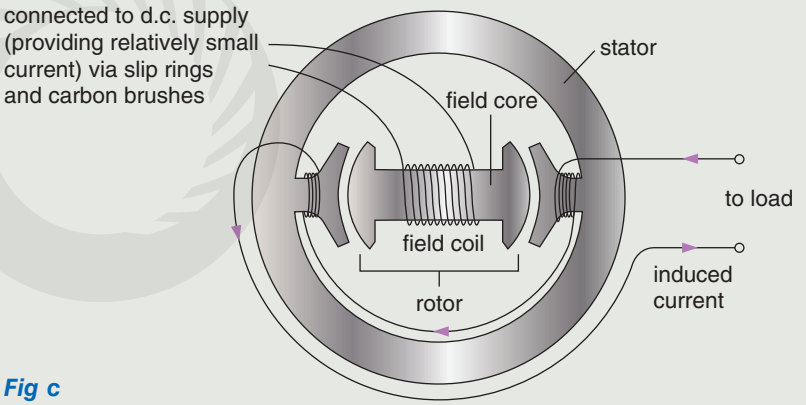



*Fig c*

rotor 轉子　　stator 定子